

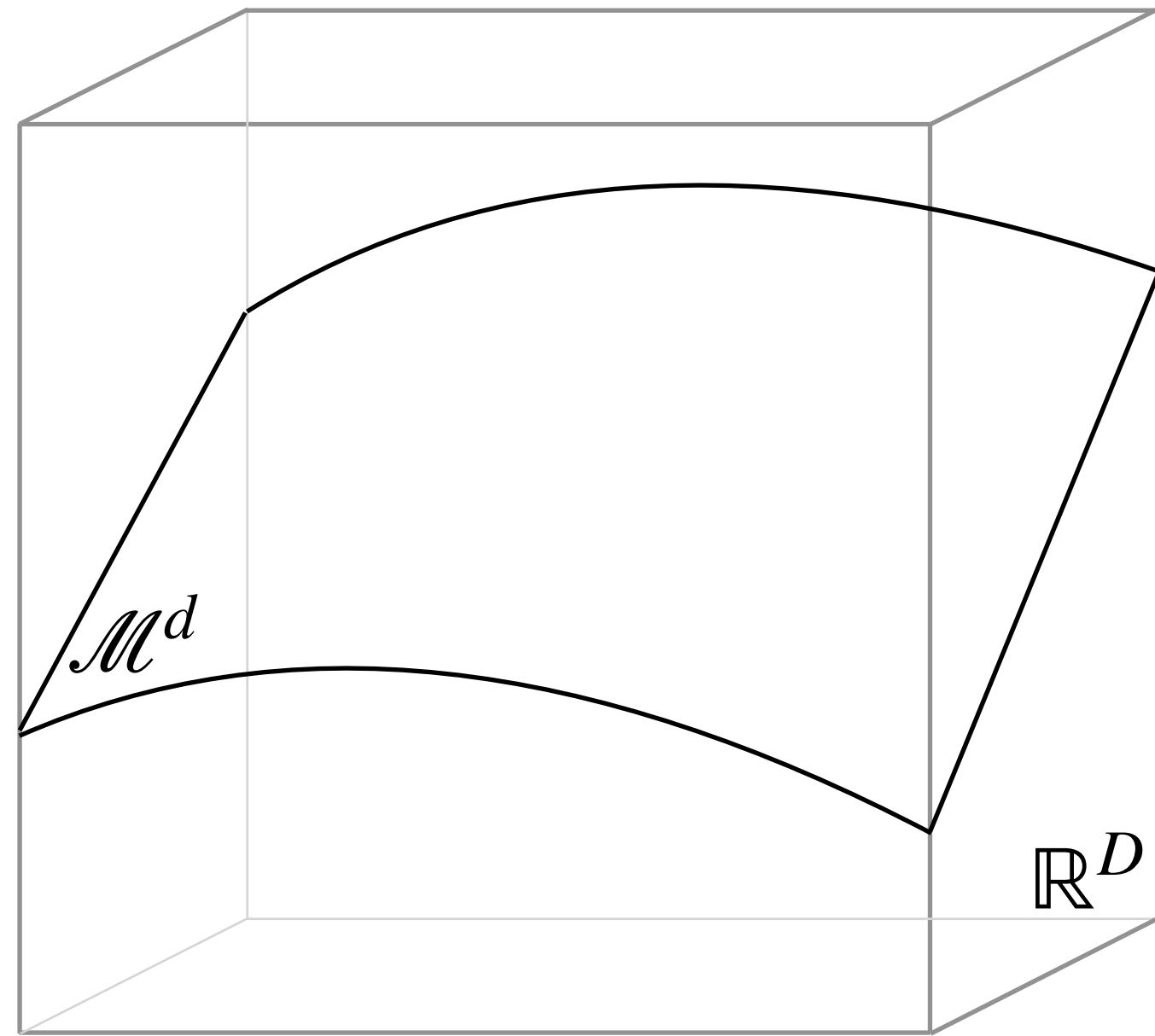
Topological Data Analysis

Lecture 13

Topological Data Analysis for Representation Learning

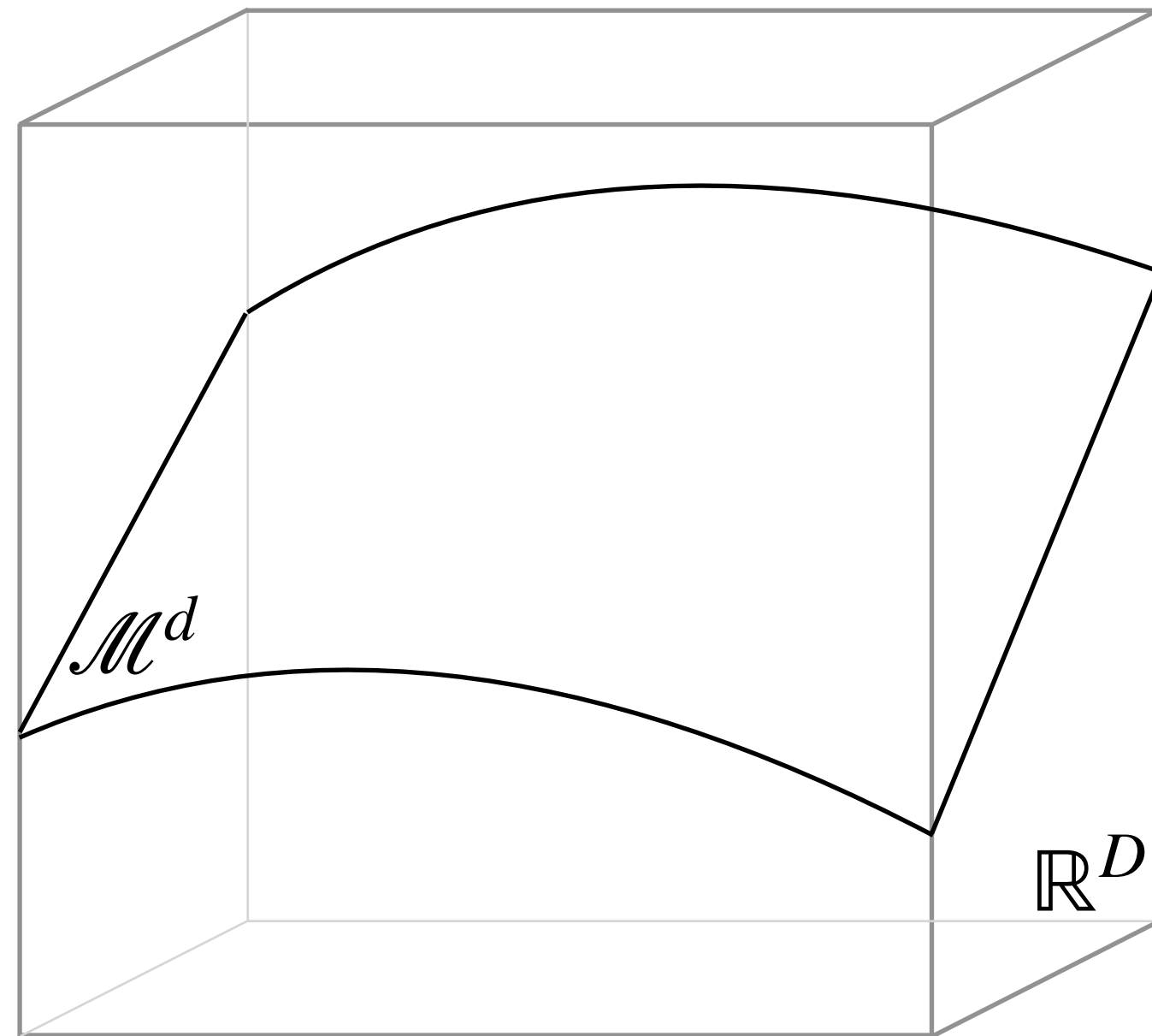
Oleg Kachan

Manifold hypothesis

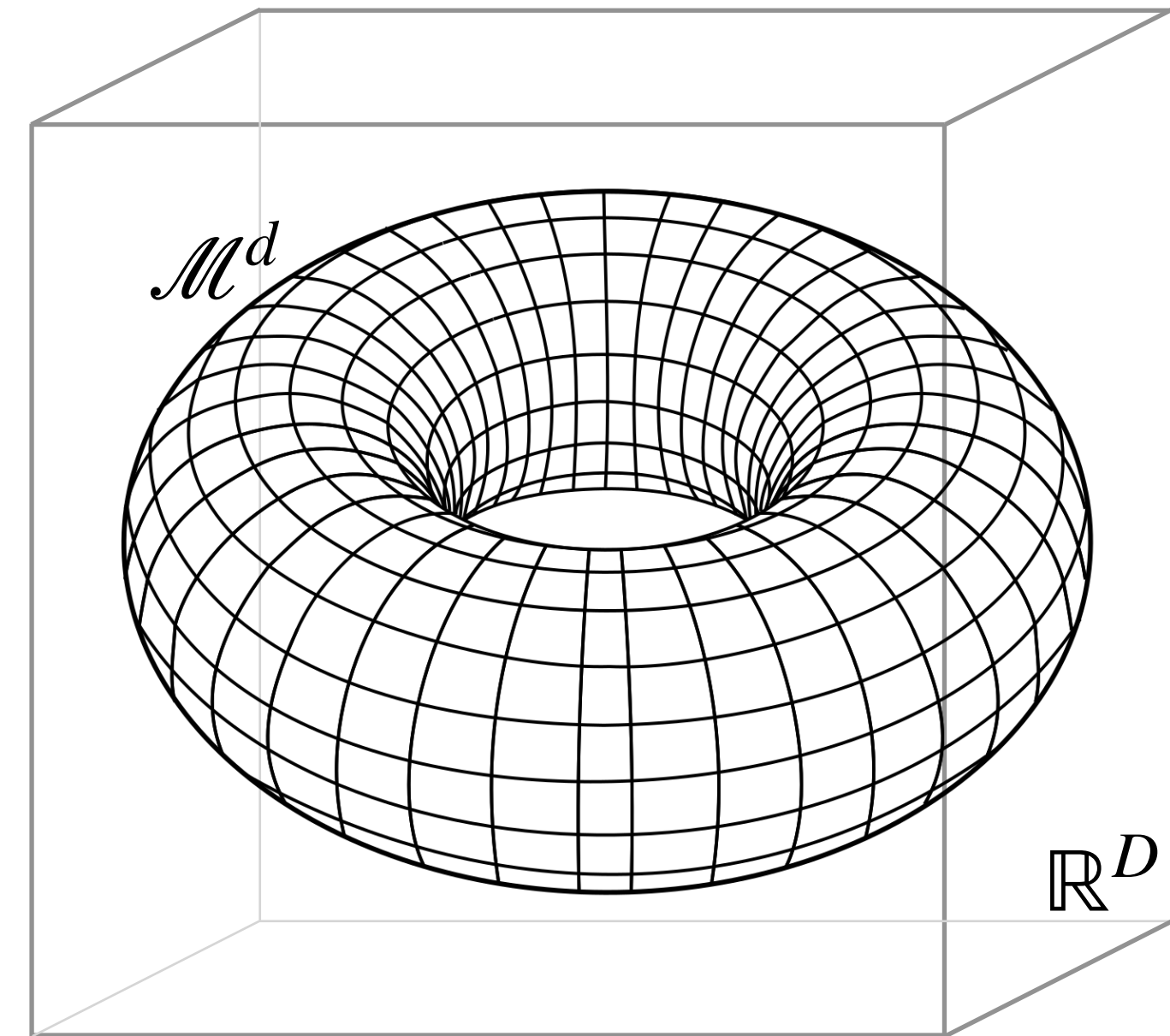


$$d \ll D$$

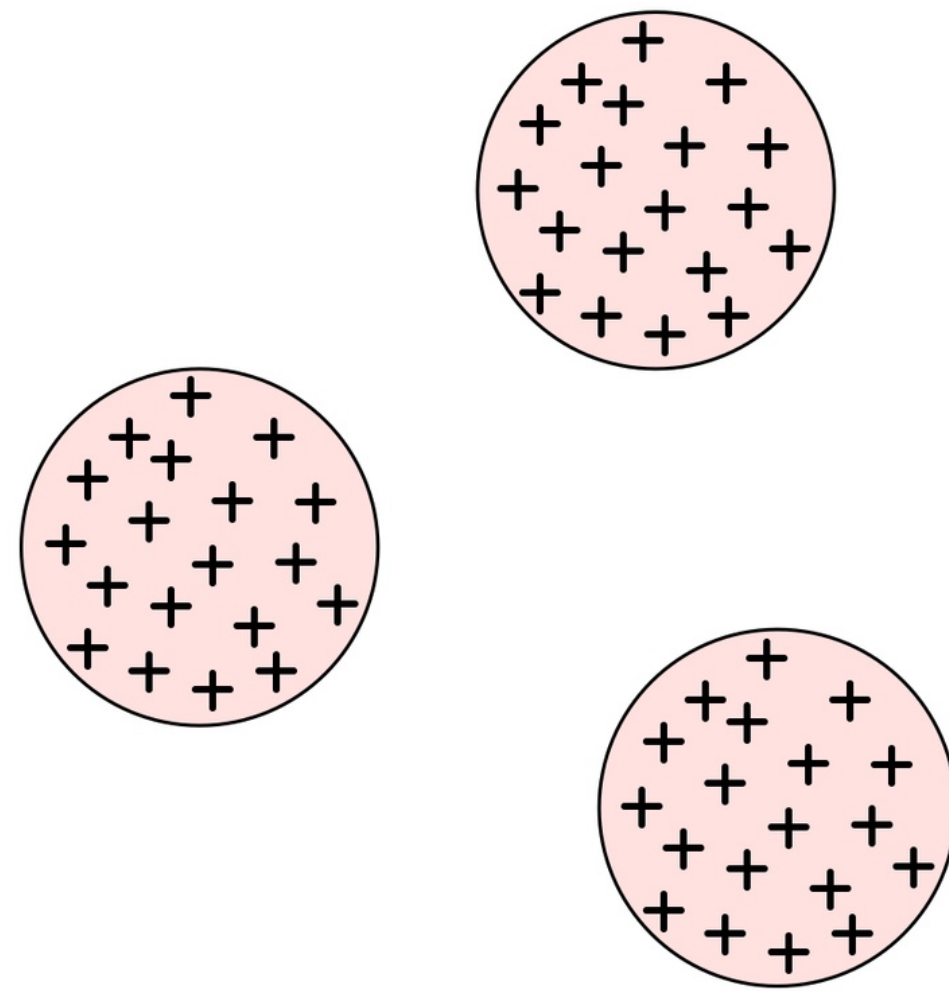
Manifold hypothesis



$$d \ll D$$



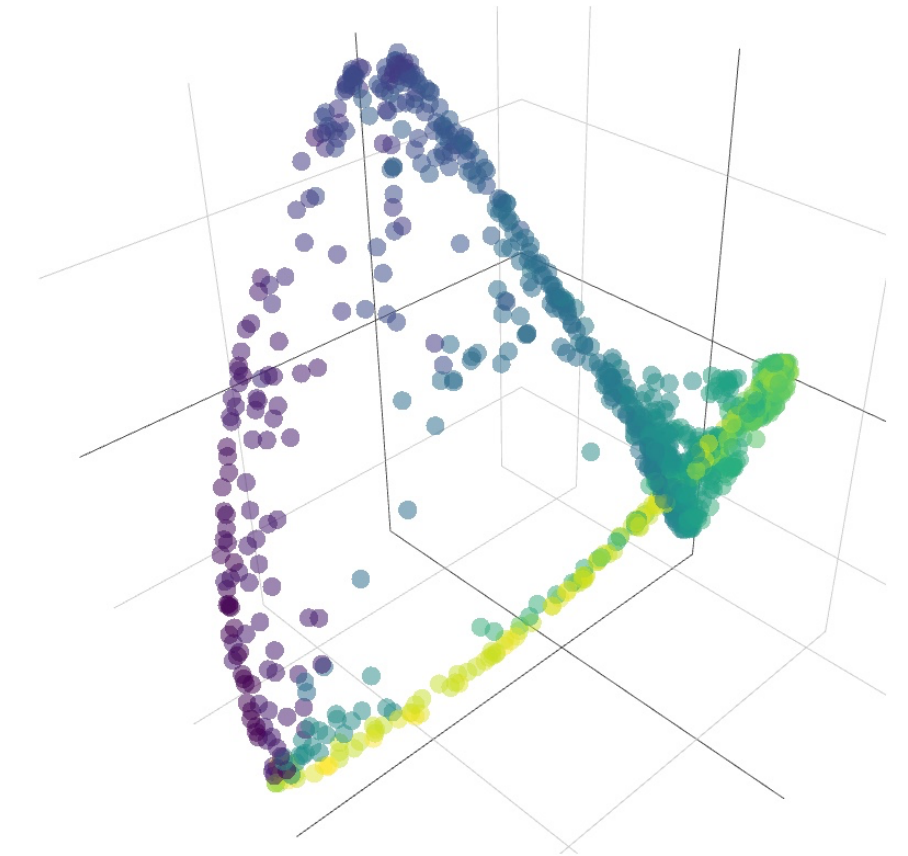
Manifold hypothesis



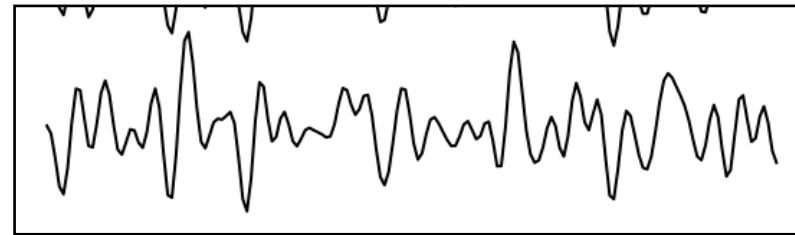
Connected components =
clusters



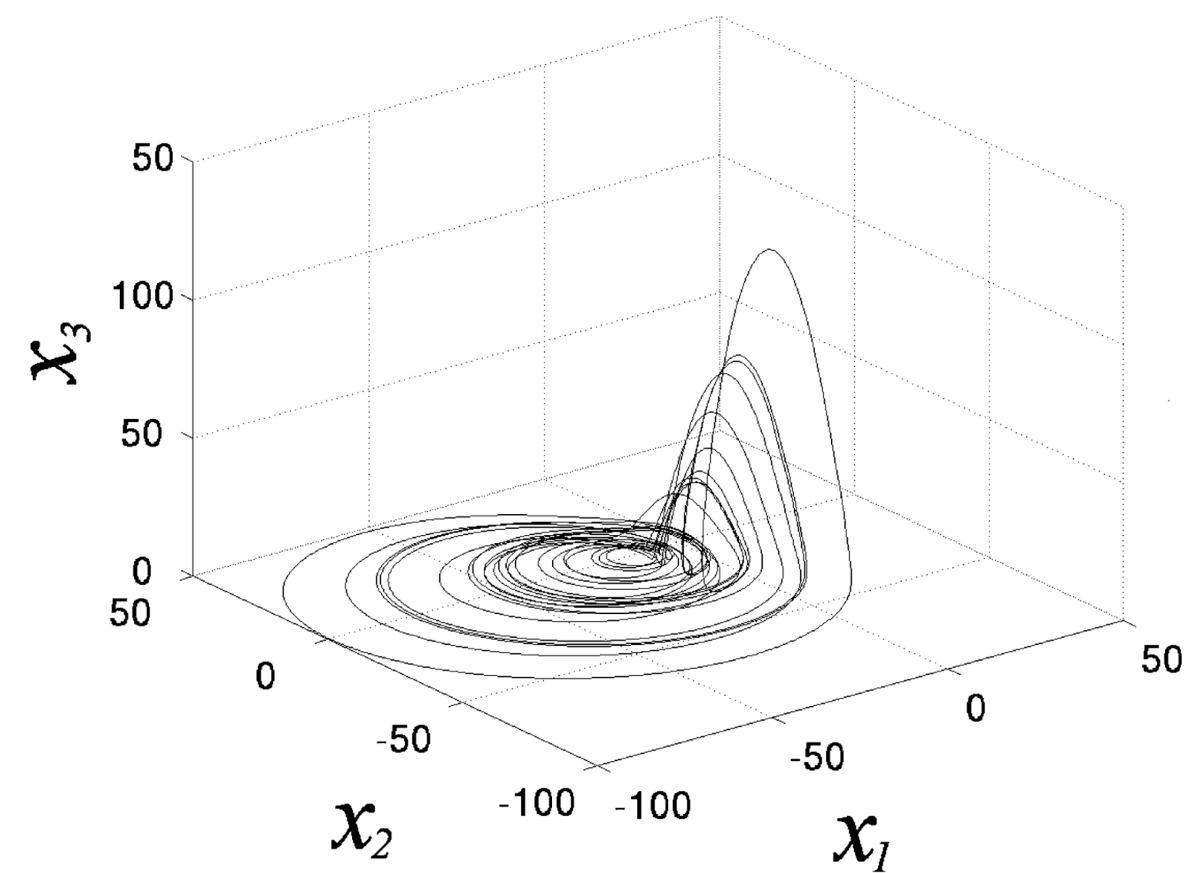
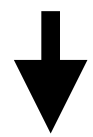
Cycles =
periodicity, regions of lower density



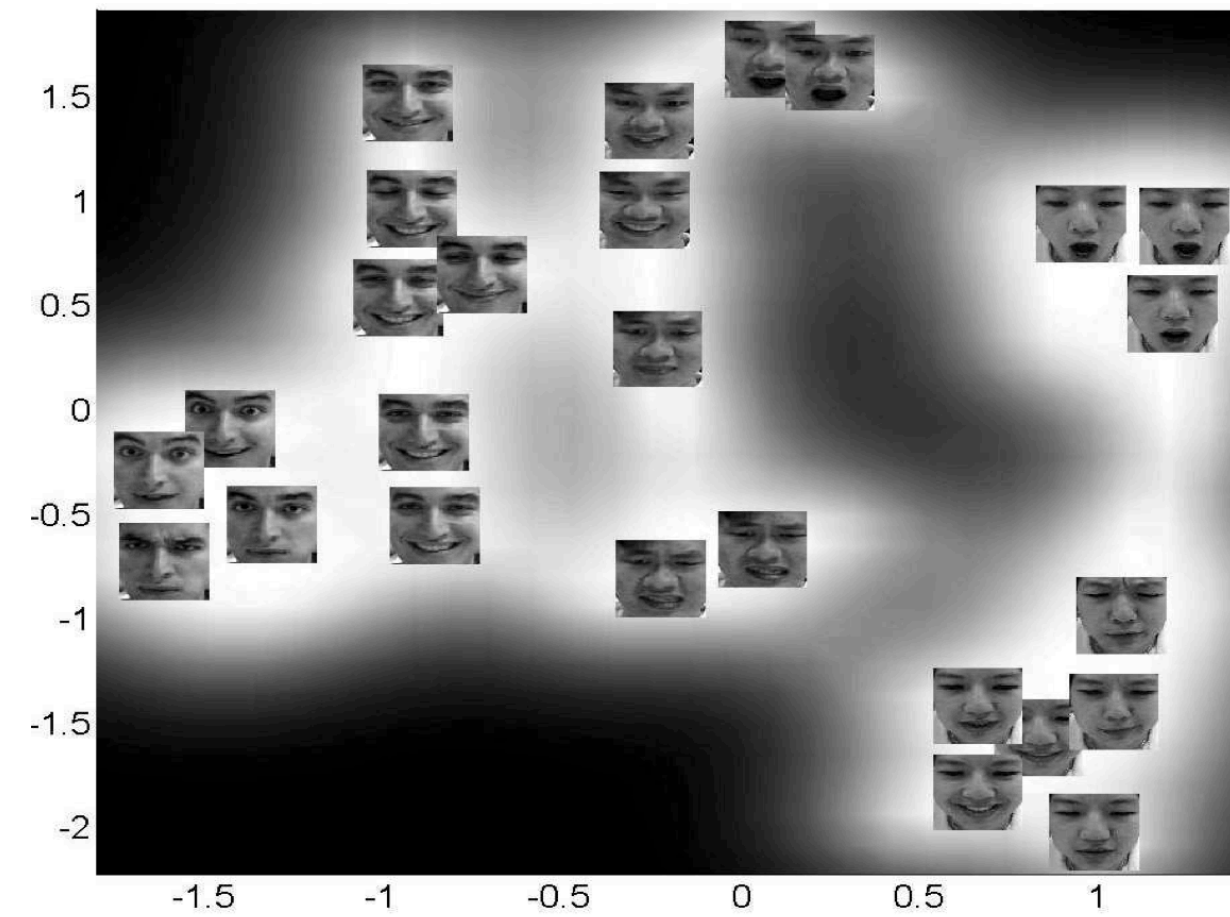
Manifold hypothesis



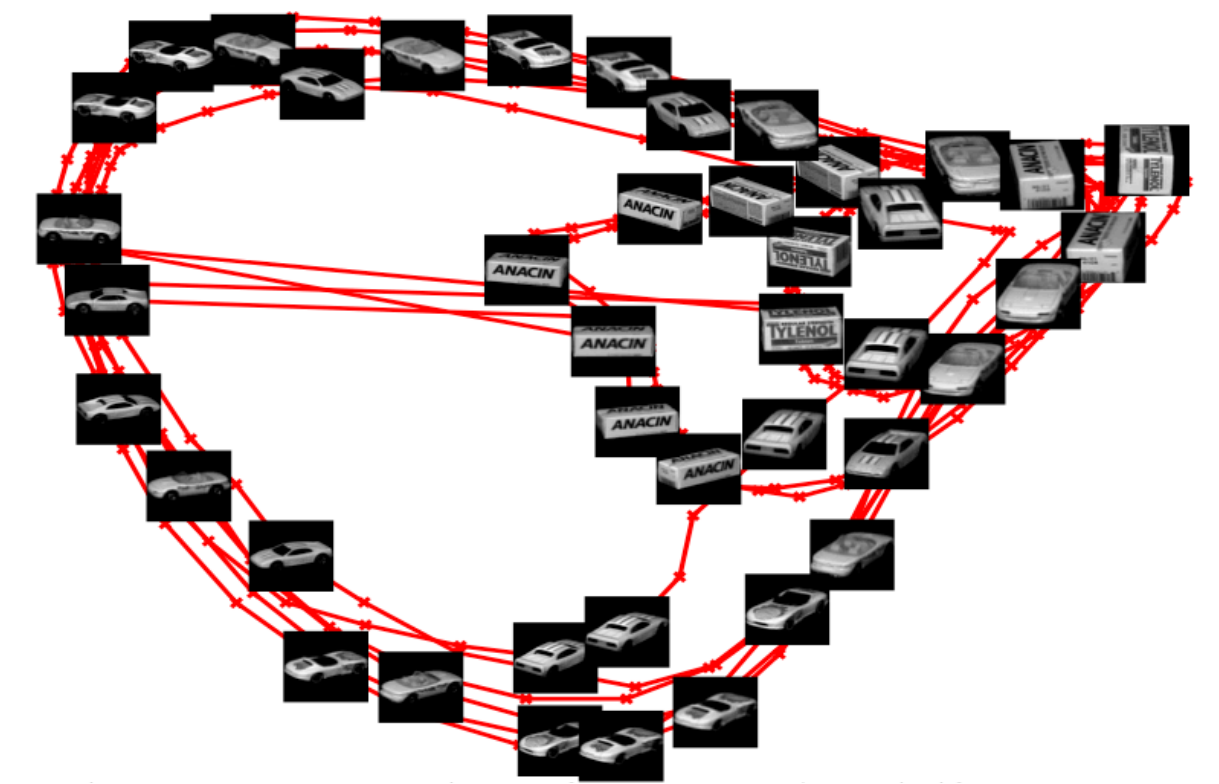
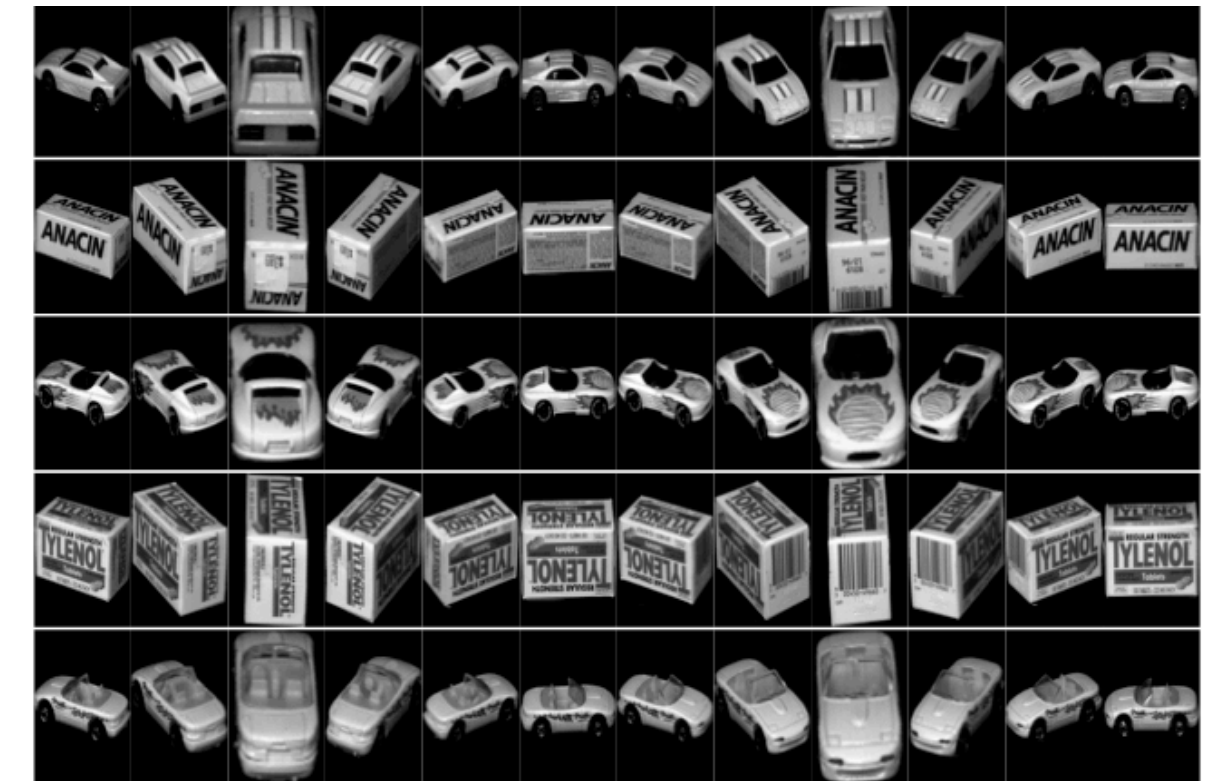
Time series



Delay embedding



COIL dataset



Dimensionality reduction

Given a dataset $X \in \mathbb{R}^D$ find a mapping $f: \mathcal{X} \in \mathbb{R}^D \rightarrow \mathcal{Z} \in \mathbb{R}^d$ where $d \ll D$, while optimizing/preserving some relevant properties of the data.

$$X \xrightarrow{f} Z$$

Properties

- variance/distances
- statistical independence

TDA for dimensionality reduction

Given a dataset $X \in \mathbb{R}^D$ find a mapping $f: \mathcal{X} \in \mathbb{R}^D \rightarrow \mathcal{Z} \in \mathbb{R}^d$ where $d \ll D$, while preserving topology of the input data.

$$X \xrightarrow{f} Z$$

$$Z \cong X$$

Properties

- variance/distances
- statistical independence
- *persistent homology groups*

TDA for representation learning

Given a dataset $X \in \mathbb{R}^D$ and mappings $f: \mathcal{X} \rightarrow \mathcal{Z}$ and $g: \mathcal{Z} \rightarrow \mathcal{Y}$
find a representation Z , close in topology to the input or output data.

$$X \xrightarrow{f} Z \xrightarrow{g} Y$$

$$Z \cong X$$

$$Z \cong Y$$

Comparing data point clouds

Persistence diagram

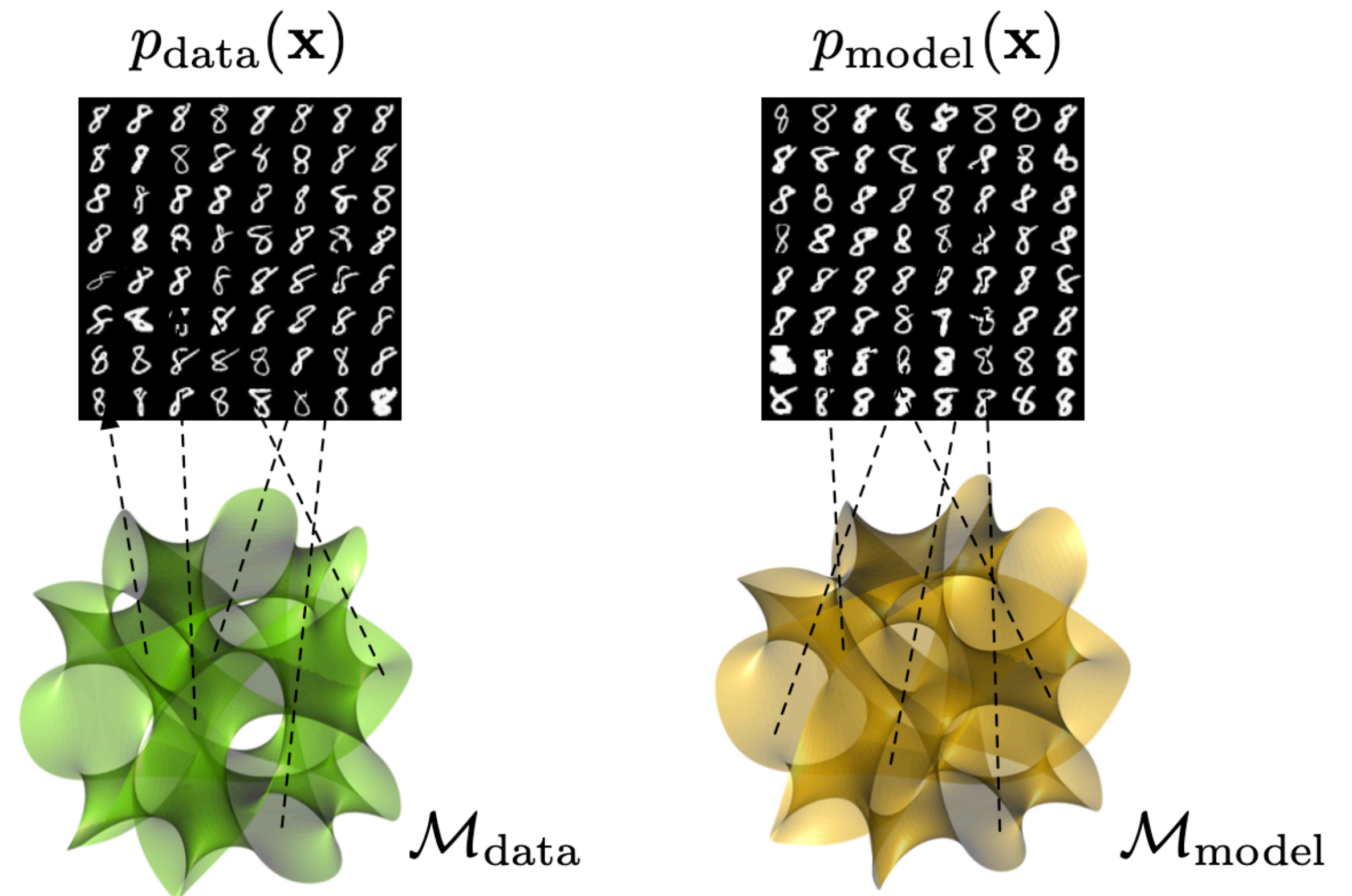
- Wasserstein distance

Persistence diagram statistics

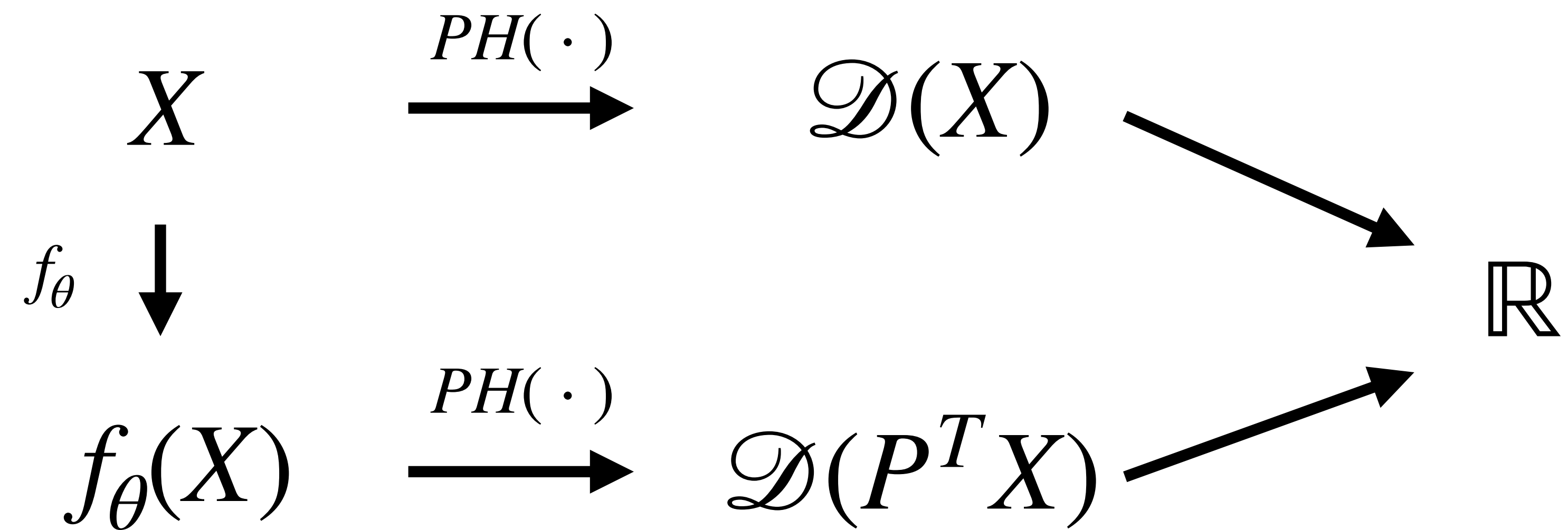
- Geometry Score (GS)

Cross-diagram

- Manifold topology divergence
- Representation topology divergence



Comparing diagrams



Comparing data point clouds

Wasserstein distance

Given a ground metric $d : X \times X \rightarrow \mathbb{R}$ optimal transport equips the space of measures $\mathcal{P}(X)$ with a metric referred to as the p -th Wasserstein distance

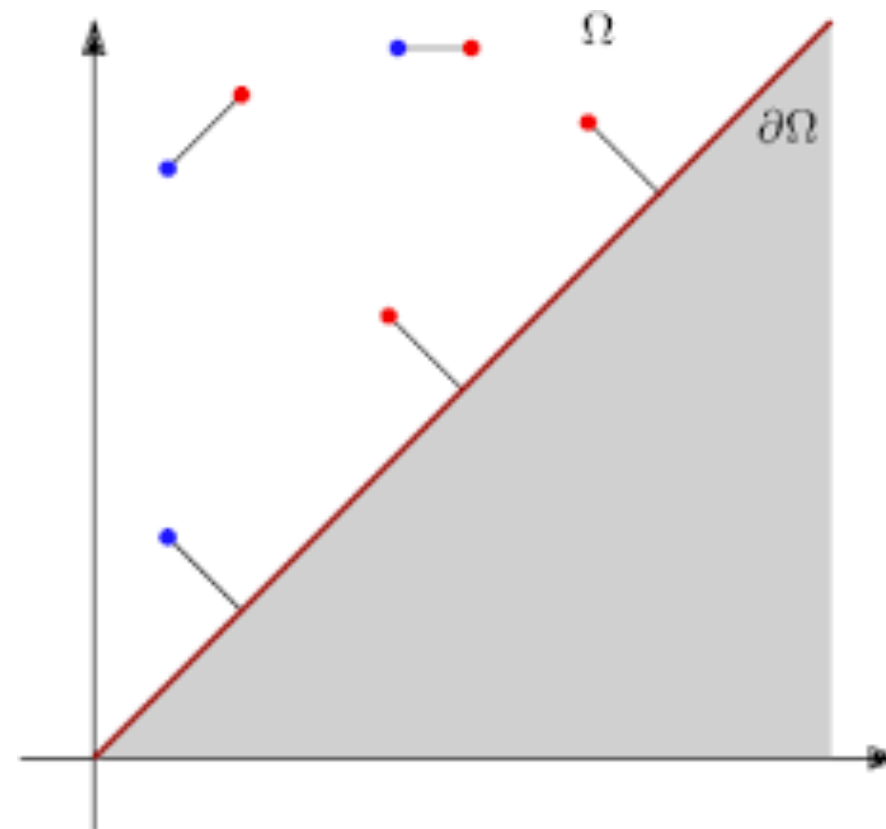
$$W_p^p(\mu, \nu) = \min_{\mathbf{T} \in \Pi(\mathbf{a}, \mathbf{b})} \langle \mathbf{T}, \tilde{\mathbf{M}} \rangle_F$$

$$\Pi(\mathbf{a}, \mathbf{b}) = \{ \mathbf{T} \in \mathbb{R}_+^{n \times m} : \mathbf{T} \mathbf{1}_m = \mathbf{a}, \mathbf{T}^T \mathbf{1}_n = \mathbf{b} \}$$

Augmented cost matrix

$$\tilde{\mathbf{M}} = \begin{pmatrix} \mathbf{M} & \Delta_{D(X)} \\ \Delta_{D(Y)} & 0 \end{pmatrix}$$

$$\begin{aligned} (\mathbf{M})_{ij} &= d^p(\mathbf{x}_i, \mathbf{y}_j) & (\Delta_{D(X)})_i &= d^p(\mathbf{x}_i, \partial\Omega) \\ & & (\Delta_{D(Y)})_j &= d^p(\mathbf{y}_j, \partial\Omega) \end{aligned}$$



Approximations

- Sinkhorn divergence, $O(n^2)$
- Sliced Wasserstein distance, $O(n \log n)$

Comparing data point clouds

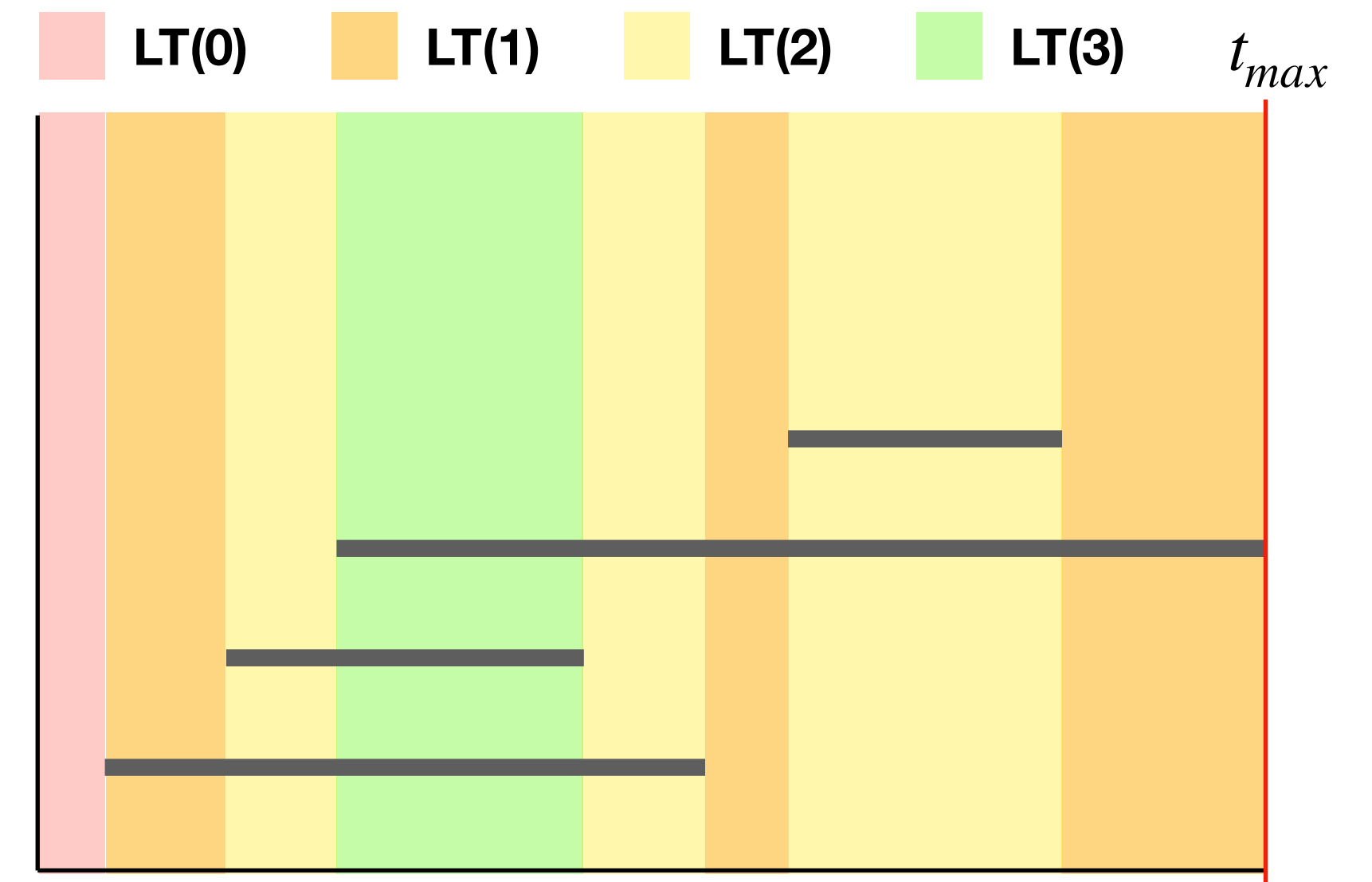
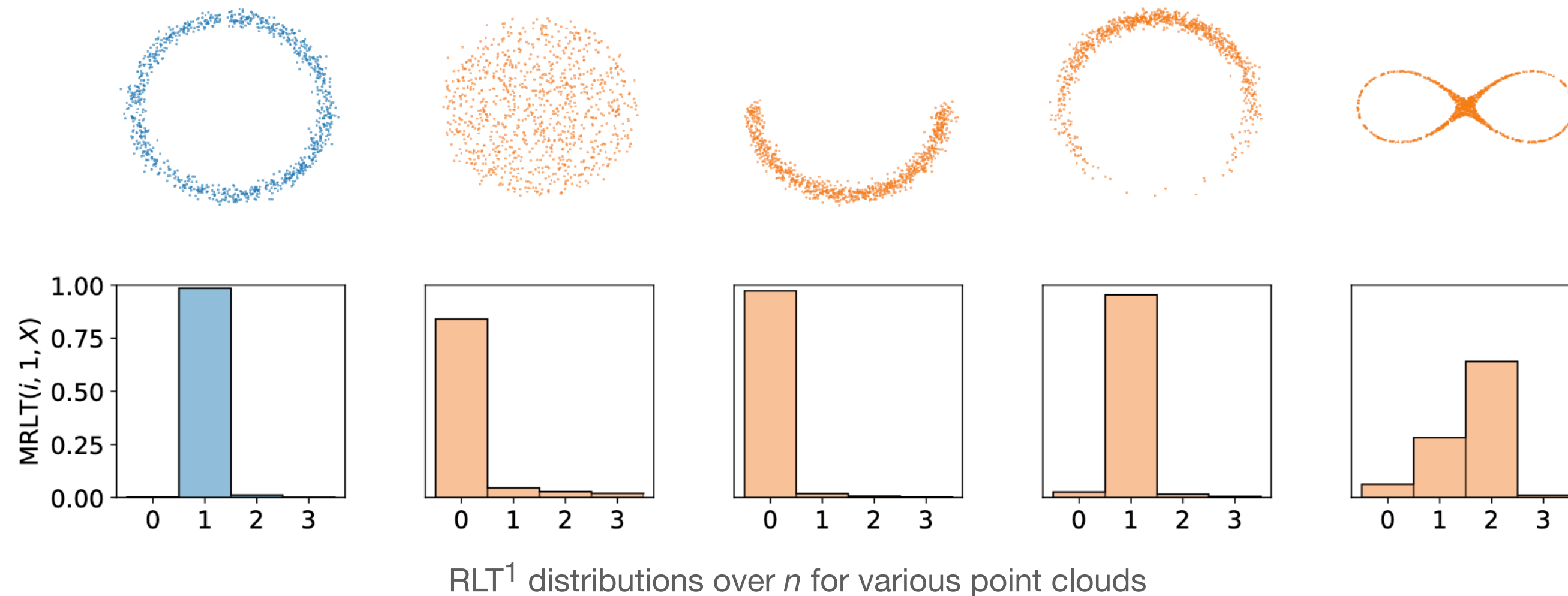
Geometry score

Given a k -th persistence diagram $D^k(X)$ the relative living time (RLT) is defined

$$\text{RLT}_X^k(n) = \frac{\mu\{t \in [0, t_{\max}] \mid \beta_k(t) = n\}}{t_{\max}}, \quad n \in \mathbb{Z}_{\geq 0}$$

Minimum enclosing radius

$$t_{\max} = \min_{x \in X} \max_{x' \in X} d(x, x')$$

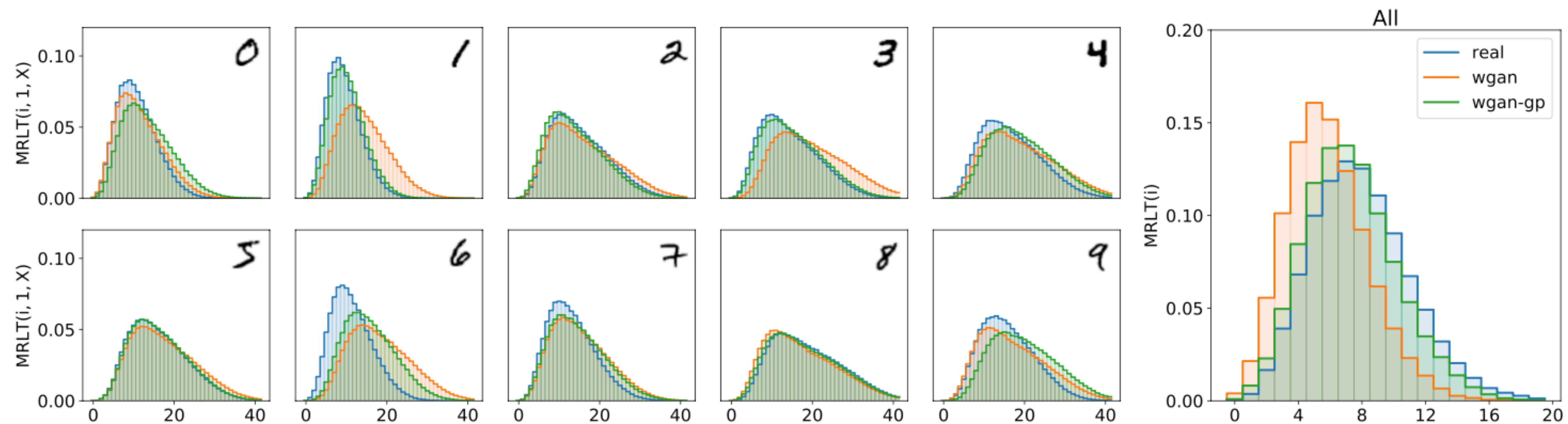


Comparing data point clouds

Geometry score

Given two point clouds X and Y the geometry score $GS(X, Y)$ is given by

$$GS(X, Y) = \sum_n^{n_{max}-1} \left(RLT_X^1(n) - RLT_Y^1(n) \right)^2$$



According to the Geometry Score WGAN-GP better fits MNIST dataset than vanilla WGAN

Comparing cross-diagrams

$$\begin{array}{ccccc} X & & & & \\ f_{\theta} \downarrow & \xrightarrow{PH(\cdot)} & \mathcal{CD}(X, f_{\theta}(X)) & \longrightarrow & \mathbb{R} \\ f_{\theta}(X) & & & & \end{array}$$

Manifold topology divergence

Cross-barcode

Given two point clouds $X, Y \in \mathcal{X}$ the cross-barcode $CB(X, Y)$ is a map to the persistent diagram of the Vietoris-Rips filtration of the union of X and Y quotiented by Y .

$$CB : (X, Y) \rightarrow D_{VR}((X \cup Y)/Y)$$

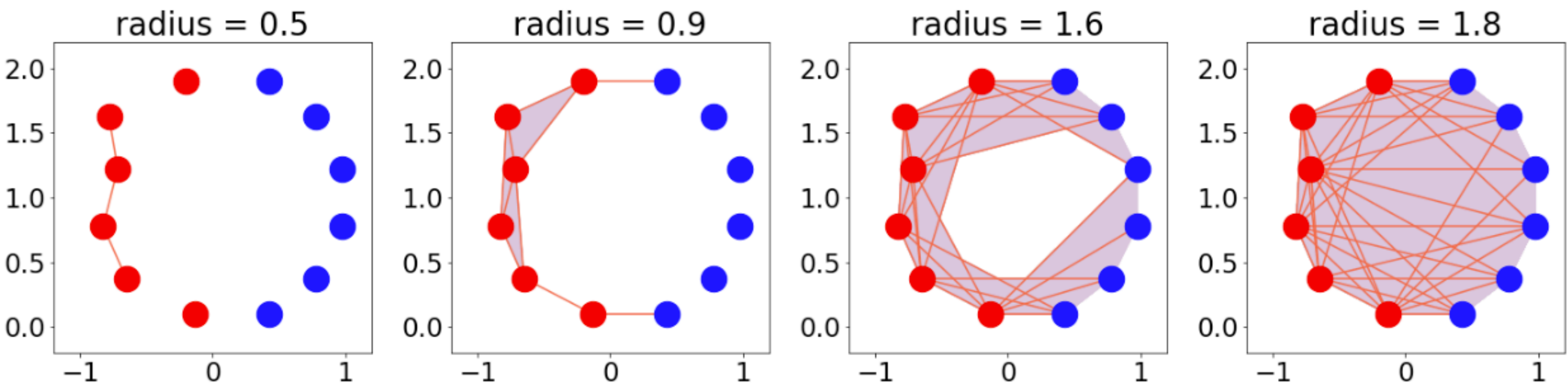
If point clouds X and Y coincide then $CB(X, Y) = \emptyset$.

$$\mathbf{A}_{XY} =$$

	x ₁	x ₂	x ₃	y ₁	y ₂
x ₁					
x ₂					
x ₃					
y ₁				0	0
y ₂				0	0

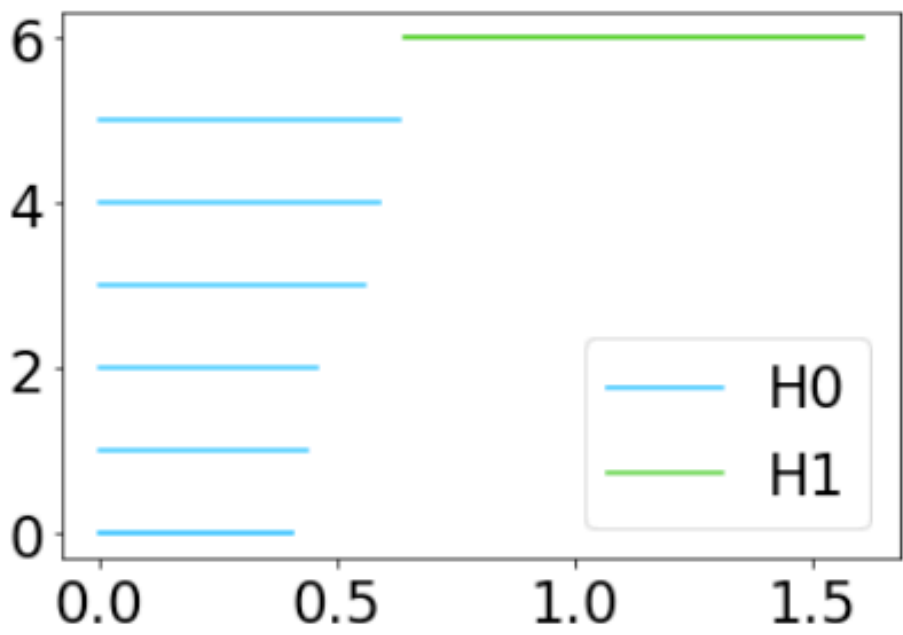
$$a_{ij} = d(x_i, x_j), d(x_i, y_j), d(x_j, y_i)$$

$$a_{ij} = 0, \quad \forall (y_i, y_j)$$



All simplices spanned by Y are born at 0 and not shown for perception ease.

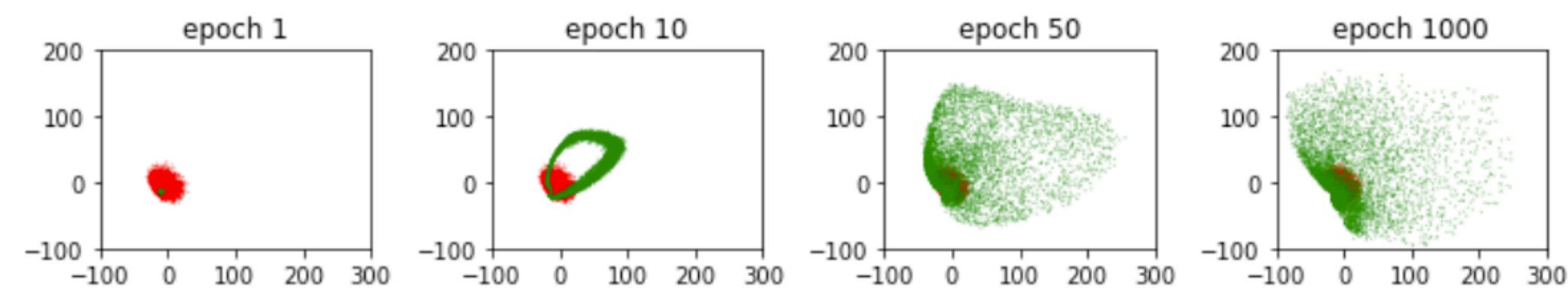
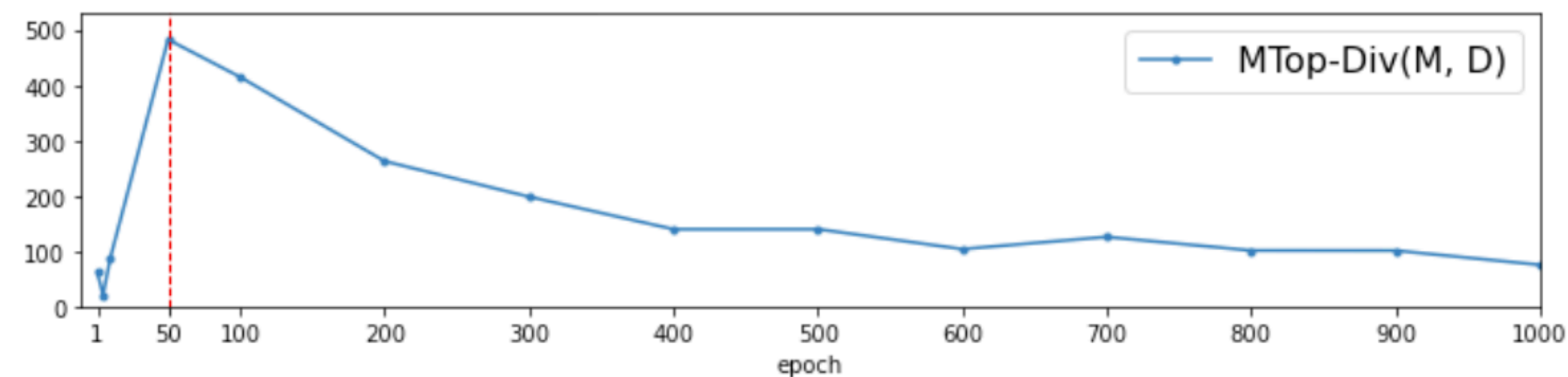
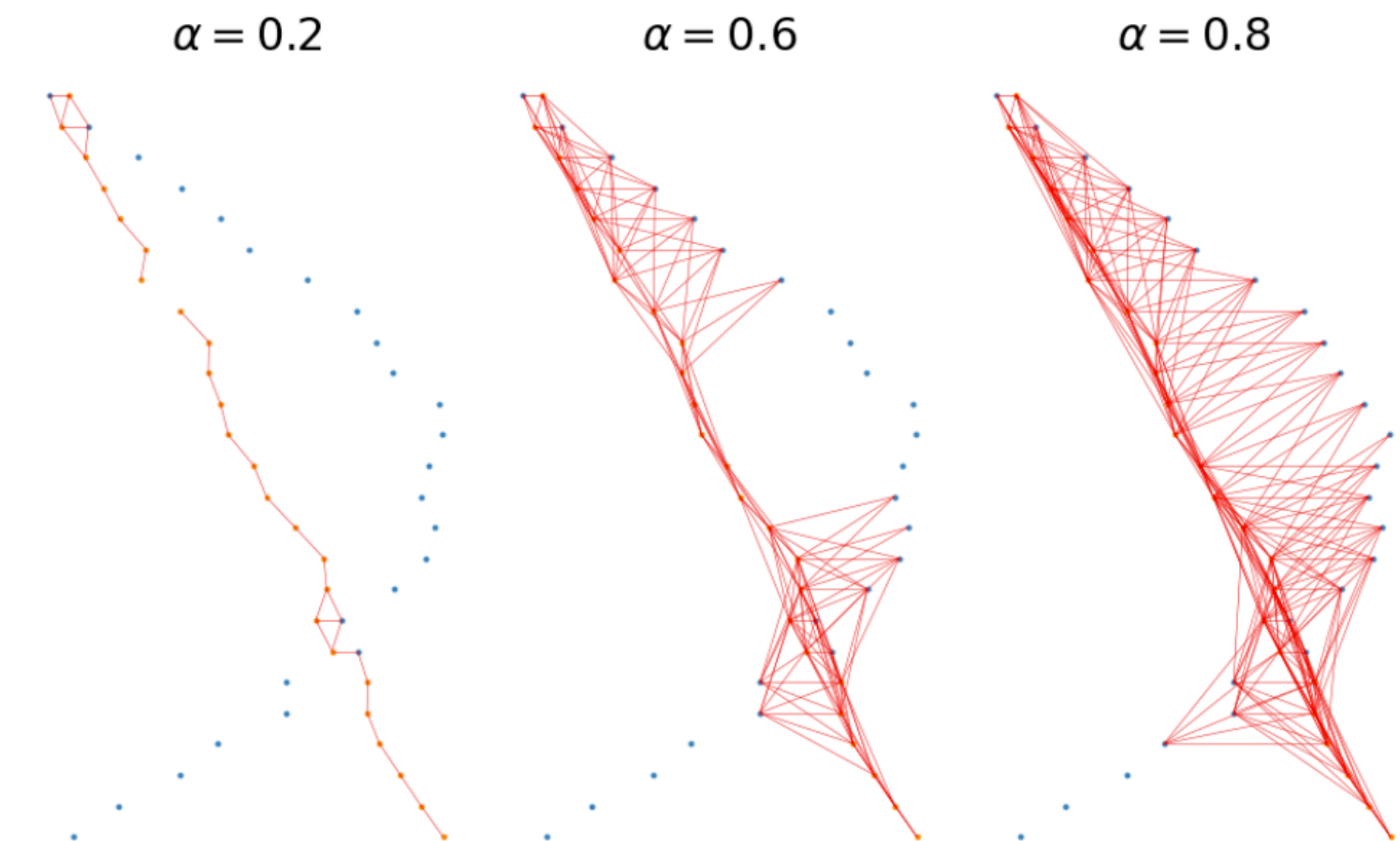
Cross-barcode (X,Y)



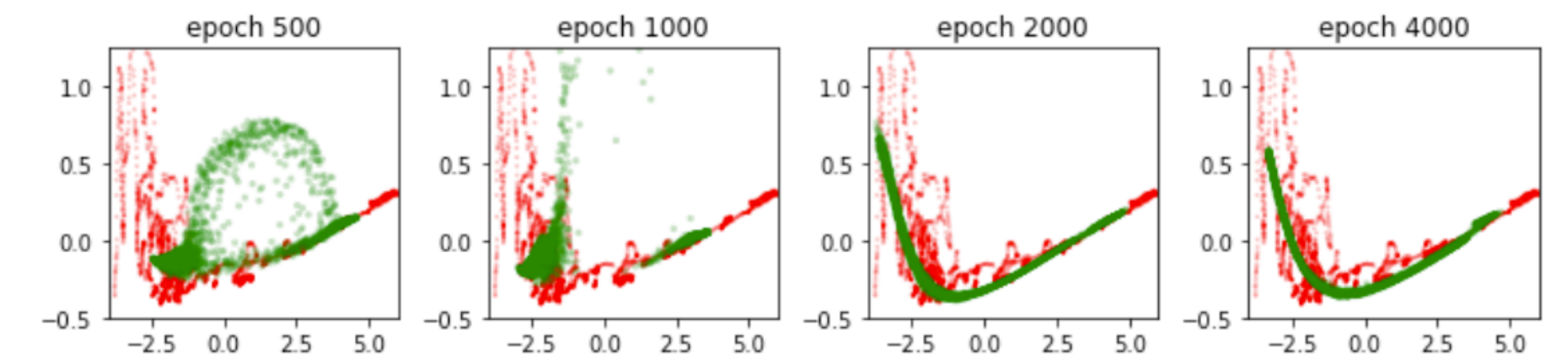
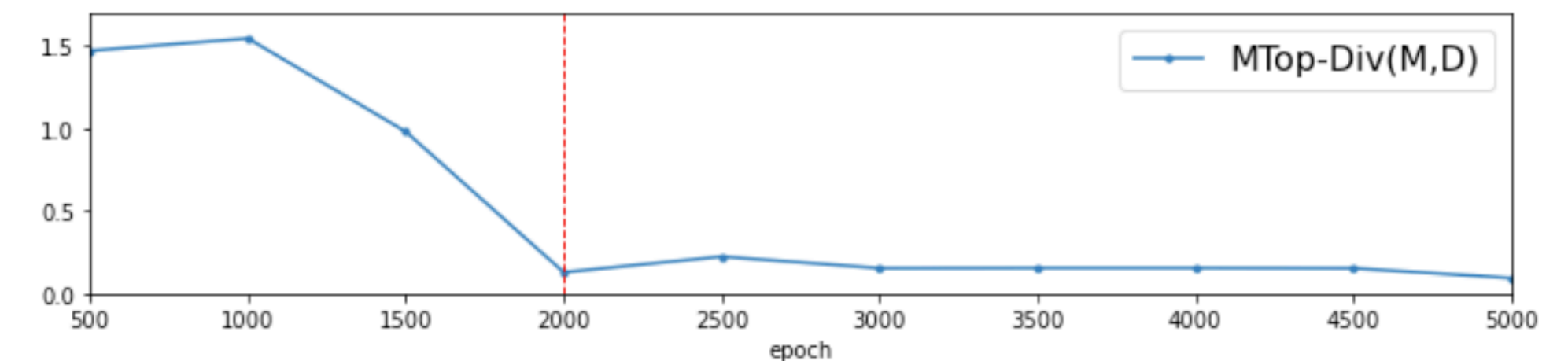
Manifold topology divergence

Given two point clouds $X, Y \in \mathcal{X}$ the manifold topology divergence $\text{MTop} - \text{Div}(X, Y)$ is given by the sum of persistences in $CB(X, Y)$

$$\text{MTop} - \text{Div}(X, Y) = \sum_i^{|CB(X,Y)|} (d_i - b_i)$$



Training dynamics of GAN applied to 3D shapes



Training dynamics of TimeGAN applied to market stock data

Comparing cross-diagrams

$$\begin{array}{ccccc} X & & & & \\ f_{\theta} \downarrow & \xrightarrow{PH(\cdot)} & \mathcal{CD}(X, f_{\theta}(X)) & \longrightarrow & \mathbb{R} \\ f_{\theta}(X) & & & & \end{array}$$

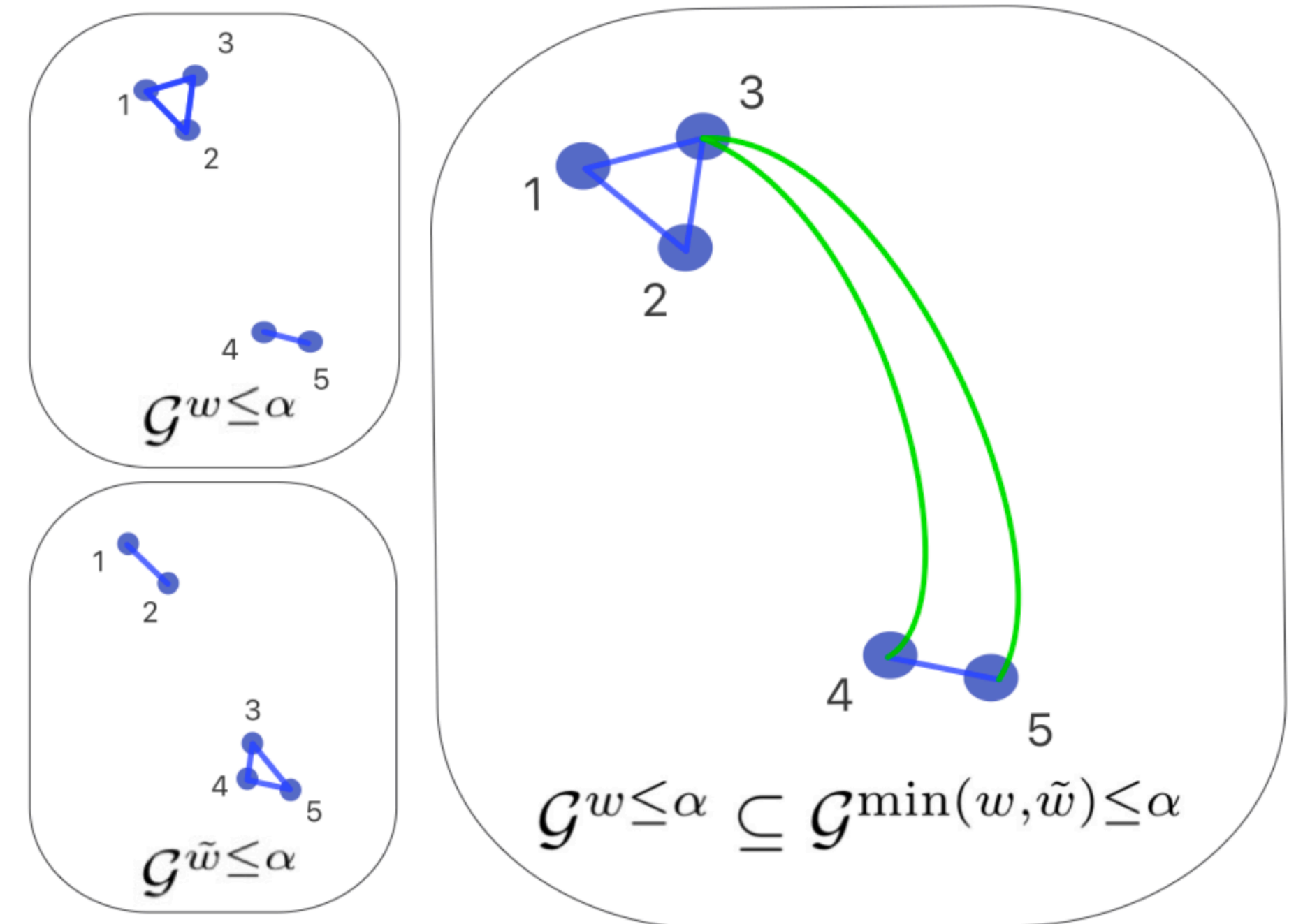
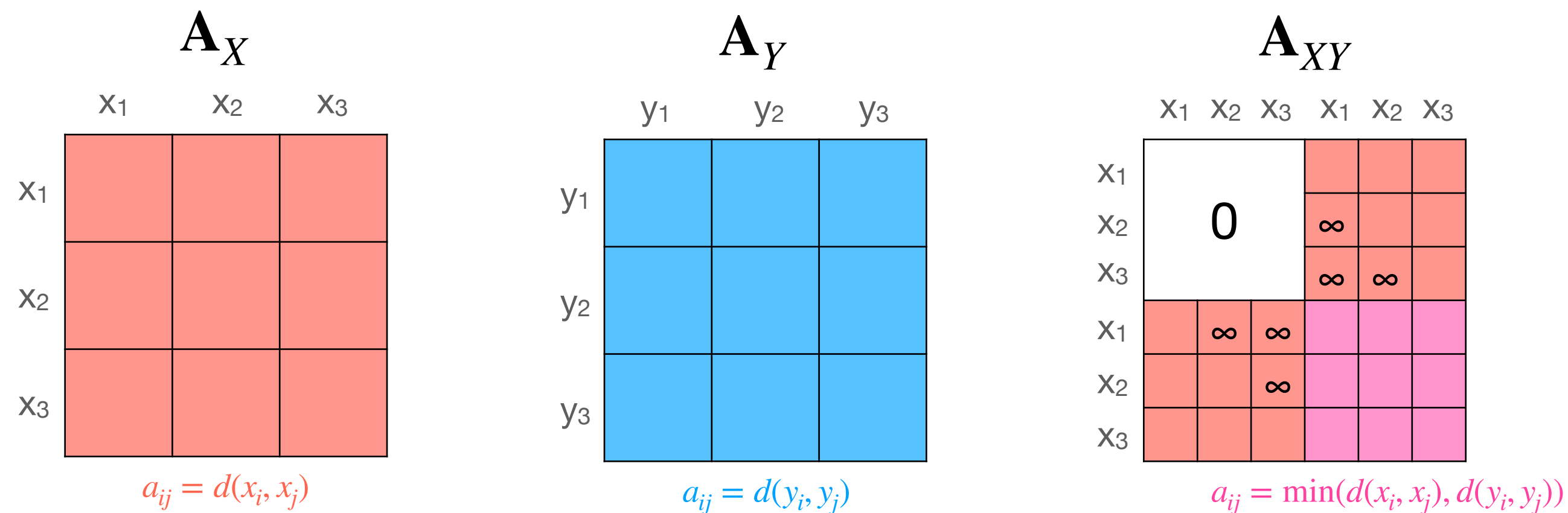
Representation topology divergence

R-cross-barcode

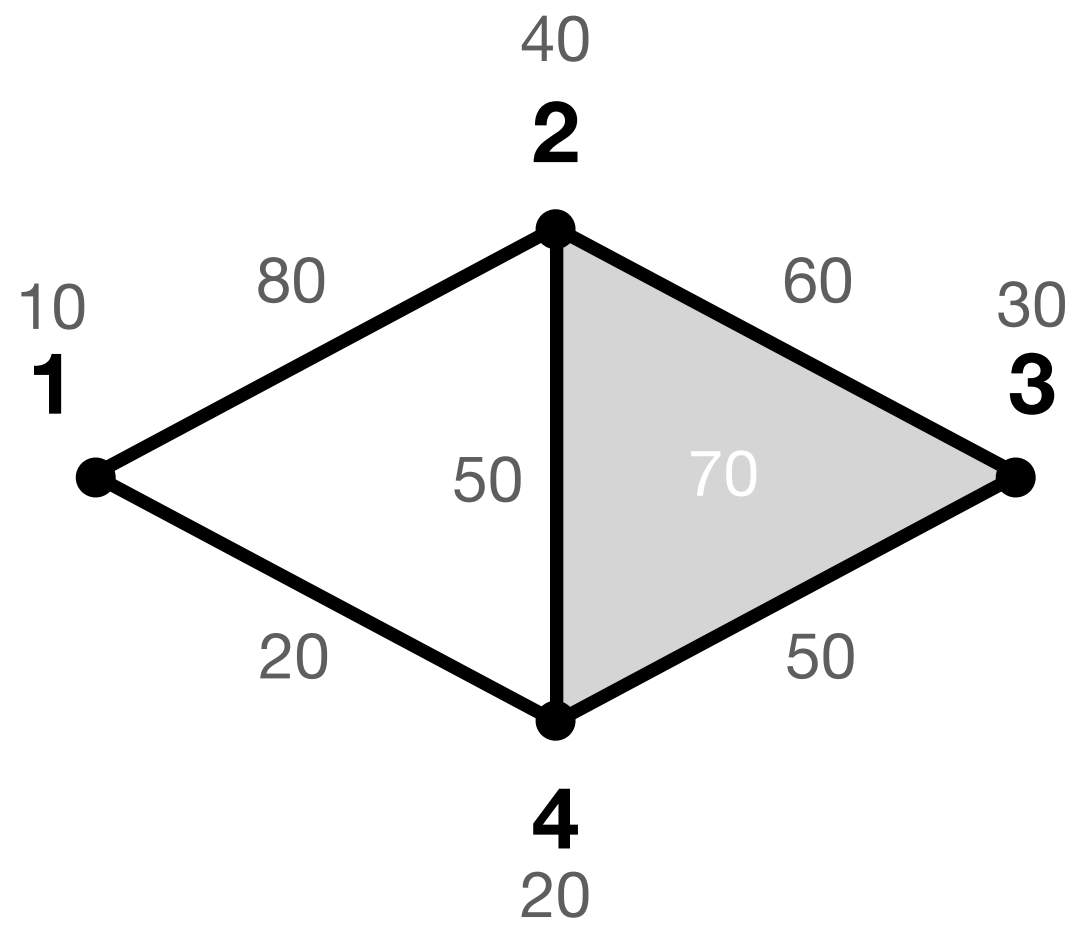
Given two point clouds $X \in \mathcal{X}, Y \in \mathcal{Y}$ with point-to-point correspondence, i.e. two embeddings of the same data, possibly in different ambient spaces, an r-cross-barcode $RCB(X, Y)$

$$RCB : (X, Y) \rightarrow D_{VR}((X, \min(X, Y)))$$

$$\text{RTop} - \text{Div}(X, Y) = \sum_i^{|RCB(X,Y)|} (d_i - b_i)$$



Differentiating persistent homology



R =

	10	20	20	30	40	50	50	60	70	80
	1	4	14	3	2	24	34	23	234	12
0										
1			1							
4			1			1	1			
14										
3							1			
2						1				
24									1	
34									1	
23									1	
12										

Persistence pairing π_X

(4, 14) 0 (1, Ø) 0
(2, 24) 0 (12, Ø) 1
(3, 34) 0
(23, 234) 1

Persistence diagram D_X

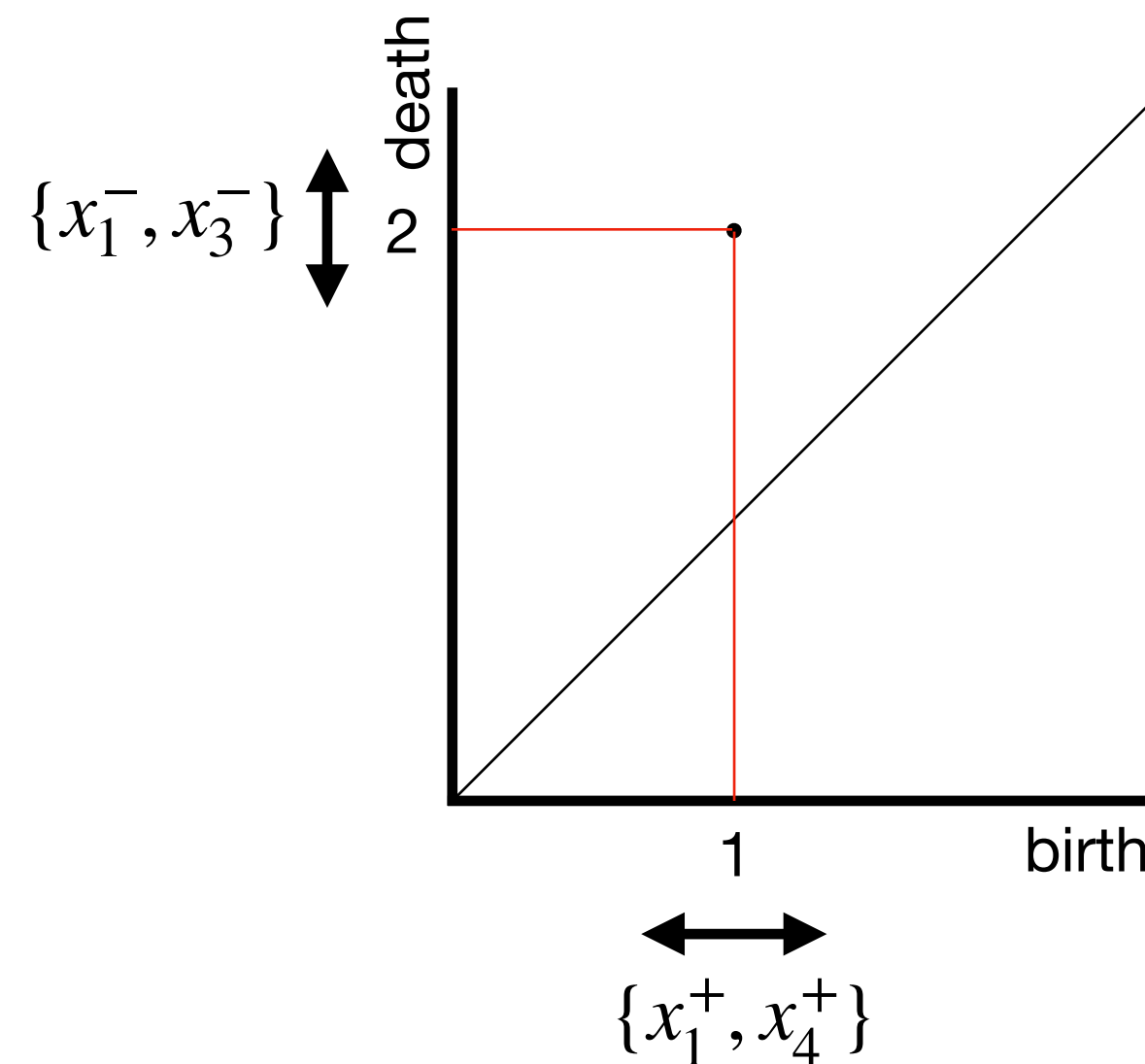
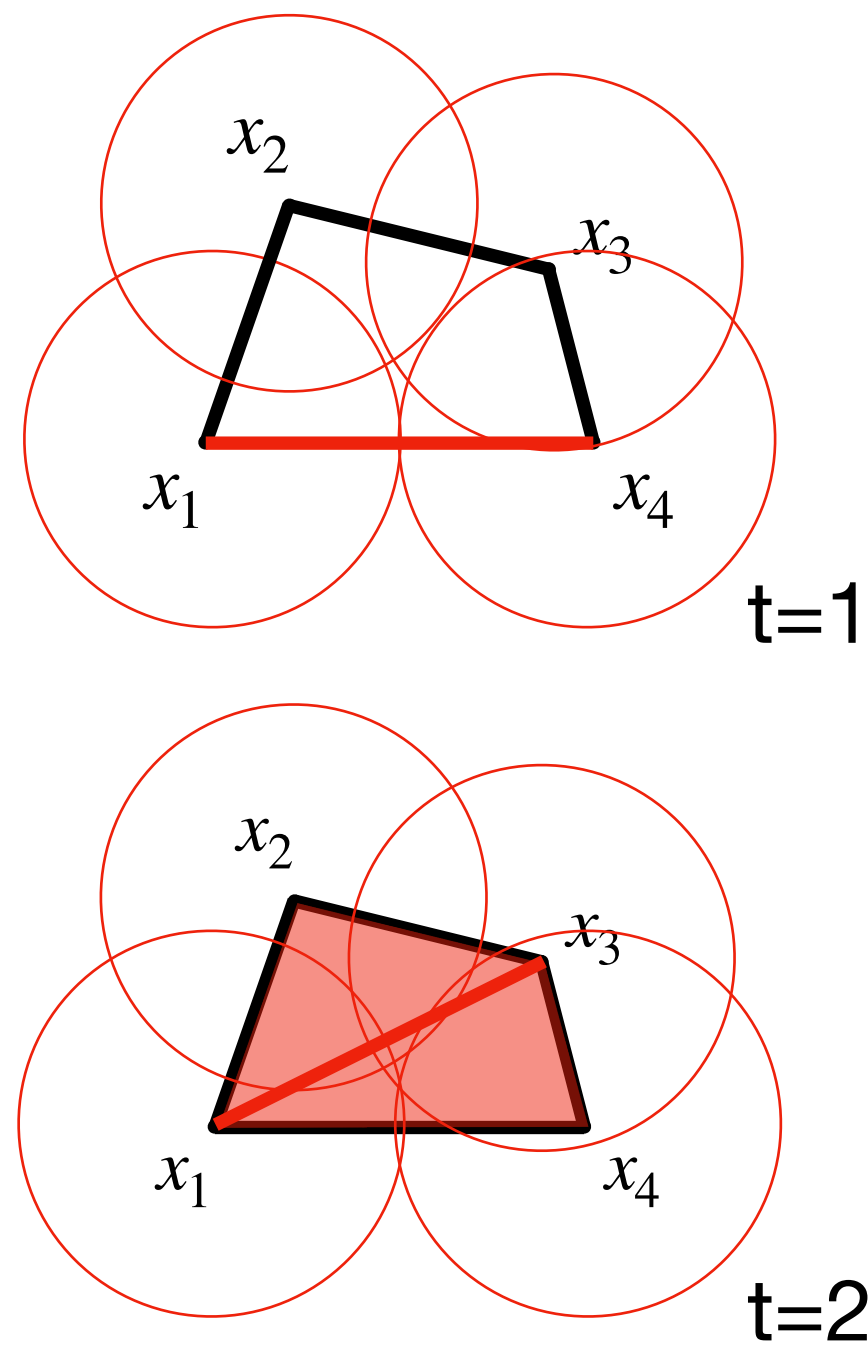
(20, 20) 0 (10, Ø) 0
(40, 50) 0 (80, Ø) 1
(30, 50) 0
(60, 70) 1

Differentiating persistent homology

Vietoris-Rips filtration

Persistent homology as a map: $PH : (X, f) \rightarrow D_f(X)$

Given a dataset X , the Vietoris-Rips simplicial complex $K_f(X)$ with the filtration function $f(\sigma) = \max_{(x_i, x_j) \in \sigma} \|x_i - x_j\|_2$ the gradient of point in the persistent diagram



Inverse function

$$\pi(\sigma) = \arg \max_{(x_i, x_j) \in \sigma} \|x_i - x_j\|_2$$

Gradient $\frac{\partial D(X)}{\partial X}$

$$\frac{\partial b_k}{\partial x_i^+} = \partial_{x_i^+} f(\sigma) = + \frac{x_i^+ - x_j^+}{\|x_i^+ - x_j^+\|_2}, \quad \frac{\partial b_k}{\partial x_j^+} = \partial_{x_j^+} f(\sigma) = - \frac{x_i^+ - x_j^+}{\|x_i^+ - x_j^+\|_2}$$

$$\frac{\partial d_k}{\partial x_i^-} = \partial_{x_i^-} f(\sigma) = + \frac{x_i^- - x_j^-}{\|x_i^- - x_j^-\|_2}, \quad \frac{\partial d_k}{\partial x_j^-} = \partial_{x_j^-} f(\sigma) = - \frac{x_i^- - x_j^-}{\|x_i^- - x_j^-\|_2}$$

Persistent homology-based projection pursuit

Given a dataset $X \in \mathbb{R}^m$ find a mapping $f: \mathcal{X} \in \mathbb{R}^m \rightarrow \mathcal{Z} \in \mathbb{R}^n$ where $n \ll m$, while optimizing/preserving some relevant properties of the data.

Properties

- variance/distances
- statistical independence
- persistent homology groups

Projection pursuit framework

- mapping in linear
- find best *index*, the value of loss function assigned for each projection Z

Persistent homology-based projection pursuit

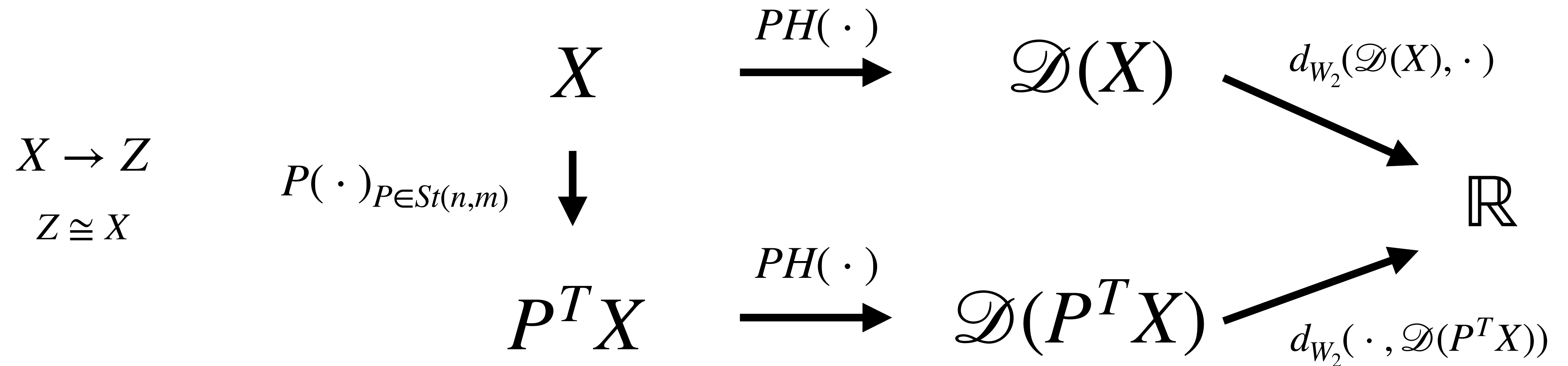
Given a dataset $X \in \mathbb{R}^m$ find a mapping $f: \mathcal{X} \in \mathbb{R}^m \rightarrow \mathcal{Z} \in \mathbb{R}^n$, where $n \ll m$, while preserving topological properties of the data in terms of persistent homology groups

$$\min_{P \in St(n,m)} W_2(\mathcal{D}(X), \mathcal{D}(P^T X))$$

Bottleneck distance bounds Gromov-Hausdorff distance

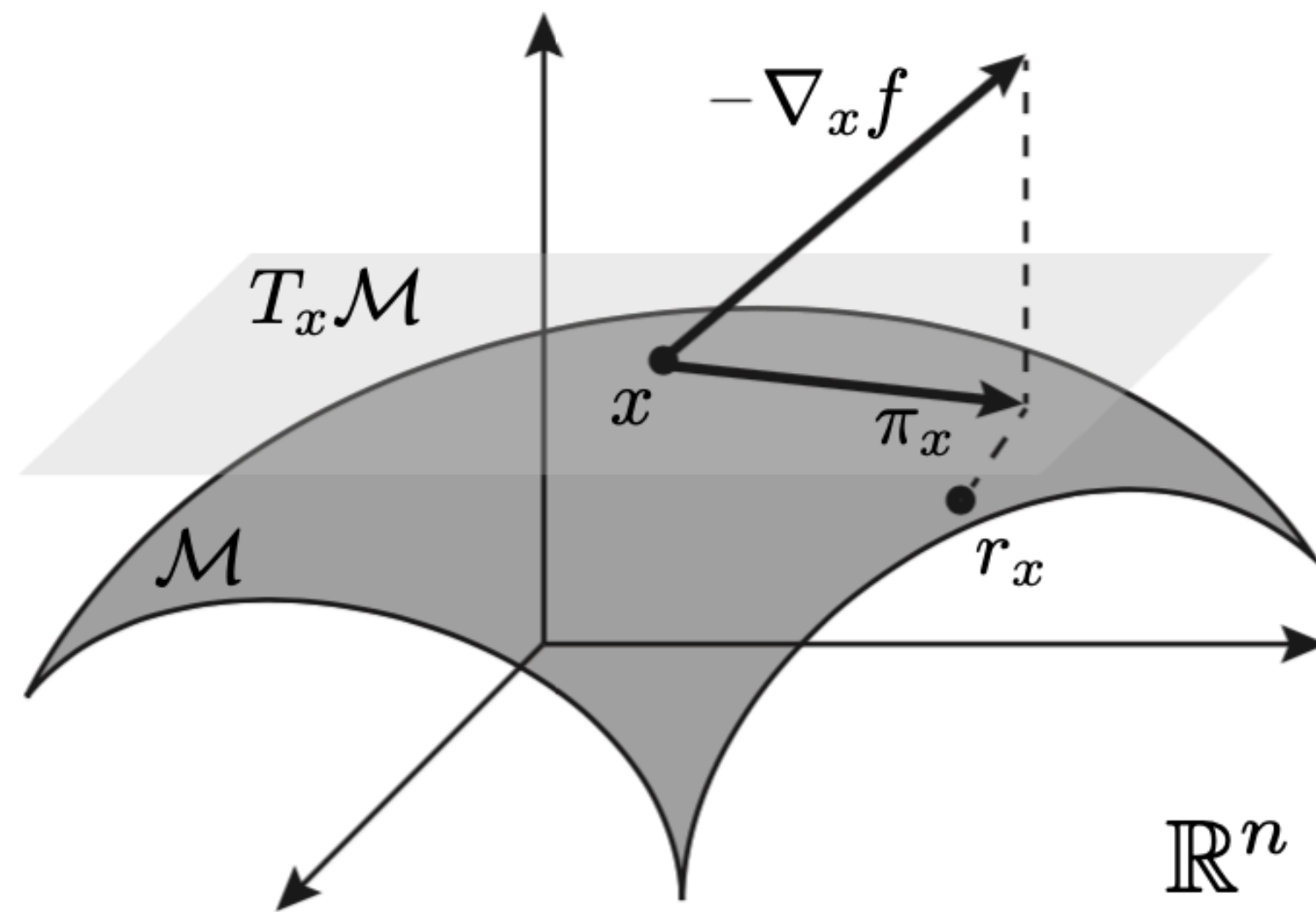
$$d_{W_\infty}(\mathcal{D}(X), \mathcal{D}(Y)) \leq d_{GH}(X, Y)$$

Persistent homology-based projection pursuit



Persistent homology-based projection pursuit

Riemannian optimization generalizes optimization algorithms to the Riemannian manifolds other than $\mathbb{R}^n$.



$$\min_{x \in St(n,m)} f(x)$$

Data: A manifold $\mathcal{M}$, a scalar field f on $\mathcal{M}$, a projection $\pi_x : \mathbb{R}^n \rightarrow T_x \mathcal{M}$, a retraction $r_x : T_x \mathcal{M} \rightarrow \mathcal{M}$, an initial iterate $x_0 \in \mathcal{M}$, step size $\alpha \in \mathbb{R}_+$

Result: Sequence of iterates $\{x_k\}$

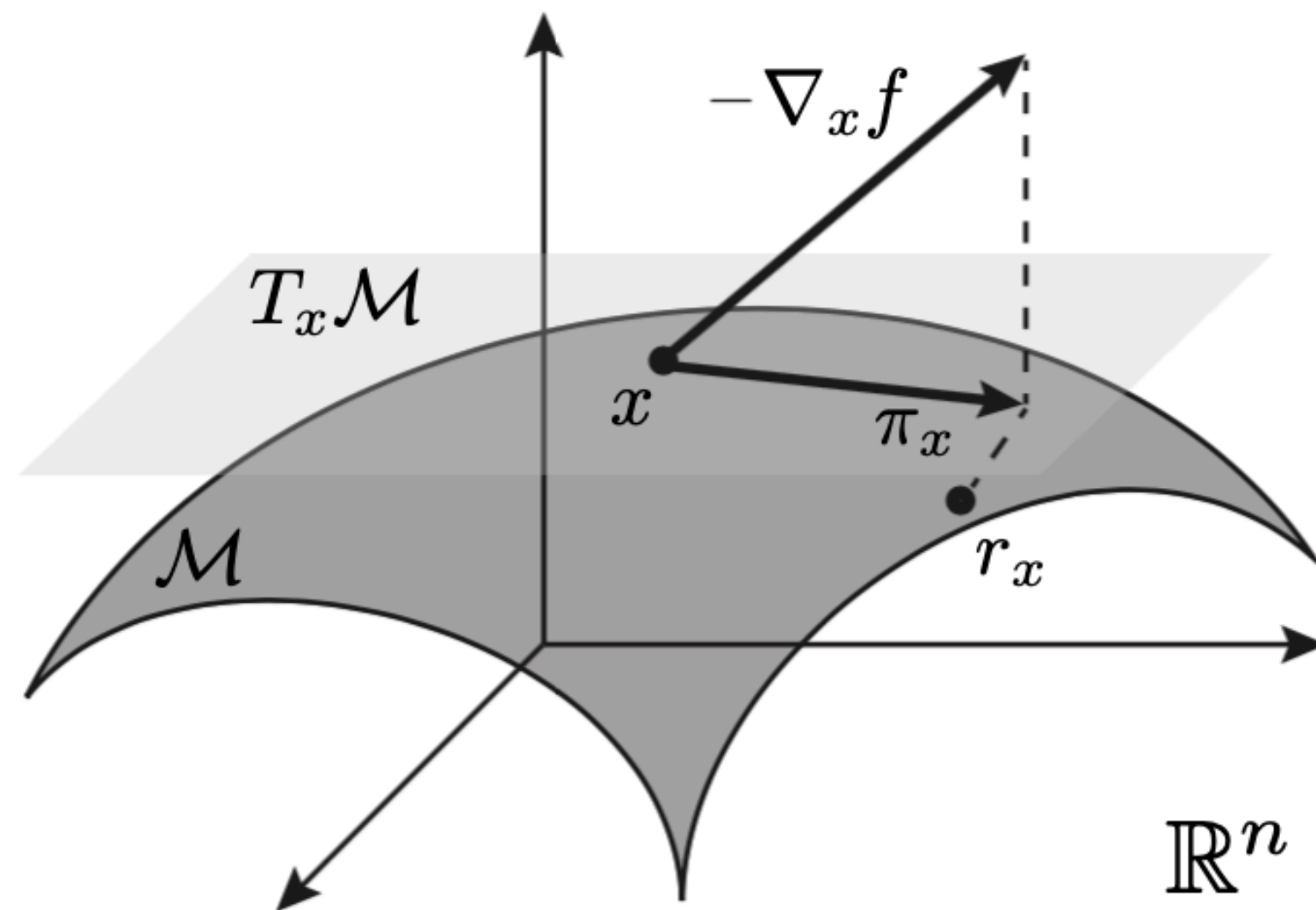
for $k = 1, 2, \dots$ **do**

$$x_{k+1} = (r_{x_k} \circ \alpha \pi_{x_k})(-\nabla_x f(x_k));$$

return $\{x_k\}$

Persistent homology-based projection pursuit

Riemannian optimization generalizes optimization algorithms to the Riemannian manifolds other than $\mathbb{R}^n$.



$$\min_{x \in St(n, m)} f(x)$$

Stiefel manifold

$$St(n, m) = \{\mathbf{X} \in \mathbb{R}^{m \times n} \mid \mathbf{X}^T \mathbf{X} = \mathbf{I}_n\}$$

Projection map

$$\pi_{\mathbf{X}} : \mathcal{R}^{m \times n} \mapsto T_{\mathbf{X}} St(n, m)$$

Let $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$, then $\pi_{\mathbf{X}}(\mathbf{A}) = \mathbf{U}\mathbf{V}^T$.

Retraction map

$$r_{\mathbf{X}} : T_{\mathbf{X}} St(n, m) \mapsto St(n, m)$$

Via polar decomp.

$$r_{\mathbf{X}}(\mathbf{V}) = (\mathbf{X} + \mathbf{V})(\mathbf{I}_p + \mathbf{V}^T \mathbf{V})^{-1/2}$$

Via QR decomp.

$$r_{\mathbf{X}}(\mathbf{V}) = qf(\mathbf{X} + \mathbf{V})$$

Persistent homology-based projection pursuit

Given a ground metric $d : X \times X \rightarrow \mathbb{R}$ optimal transport equips the space of measures $\mathcal{P}(X)$ with a metric referred to as the Wasserstein distance

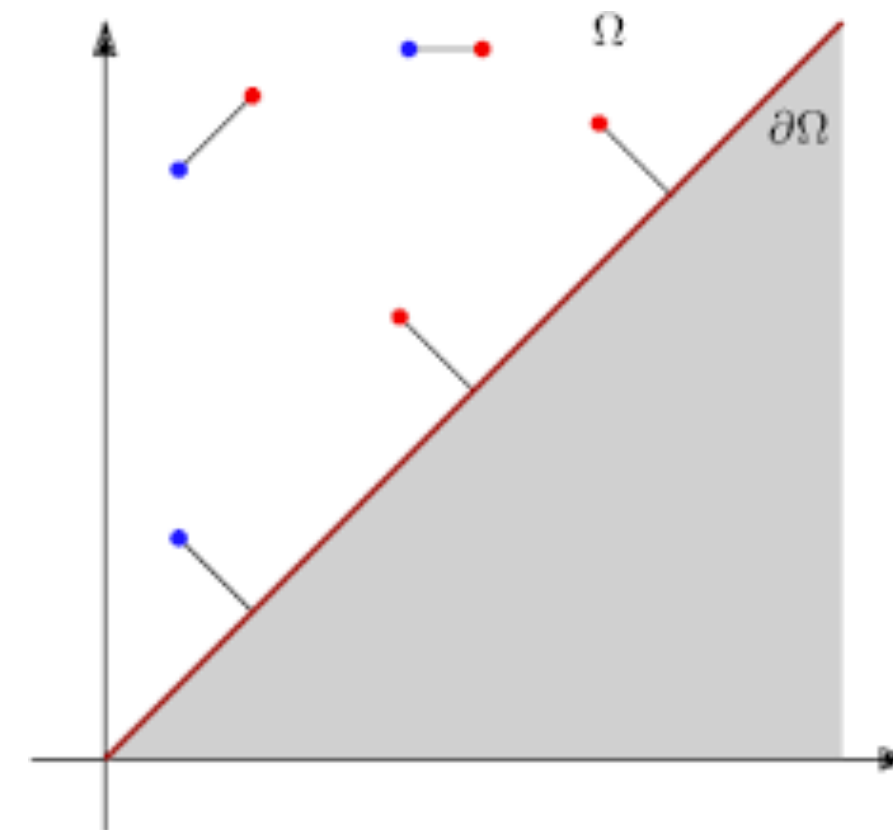
$$W_p^p(\mu, \nu) = \min_{\mathbf{T} \in \Pi(\mathbf{a}, \mathbf{b})} \langle \mathbf{T}, \tilde{\mathbf{M}} \rangle_F$$

$$\Pi(\mathbf{a}, \mathbf{b}) = \{ \mathbf{T} \in \mathbb{R}_+^{n \times m} : \mathbf{T} \mathbf{1}_m = \mathbf{a}, \mathbf{T}^T \mathbf{1}_n = \mathbf{b} \}$$

Augmented cost matrix

$$\tilde{\mathbf{M}} = \begin{pmatrix} \mathbf{M} & \Delta_{D(X)} \\ \Delta_{D(Y)} & 0 \end{pmatrix}$$

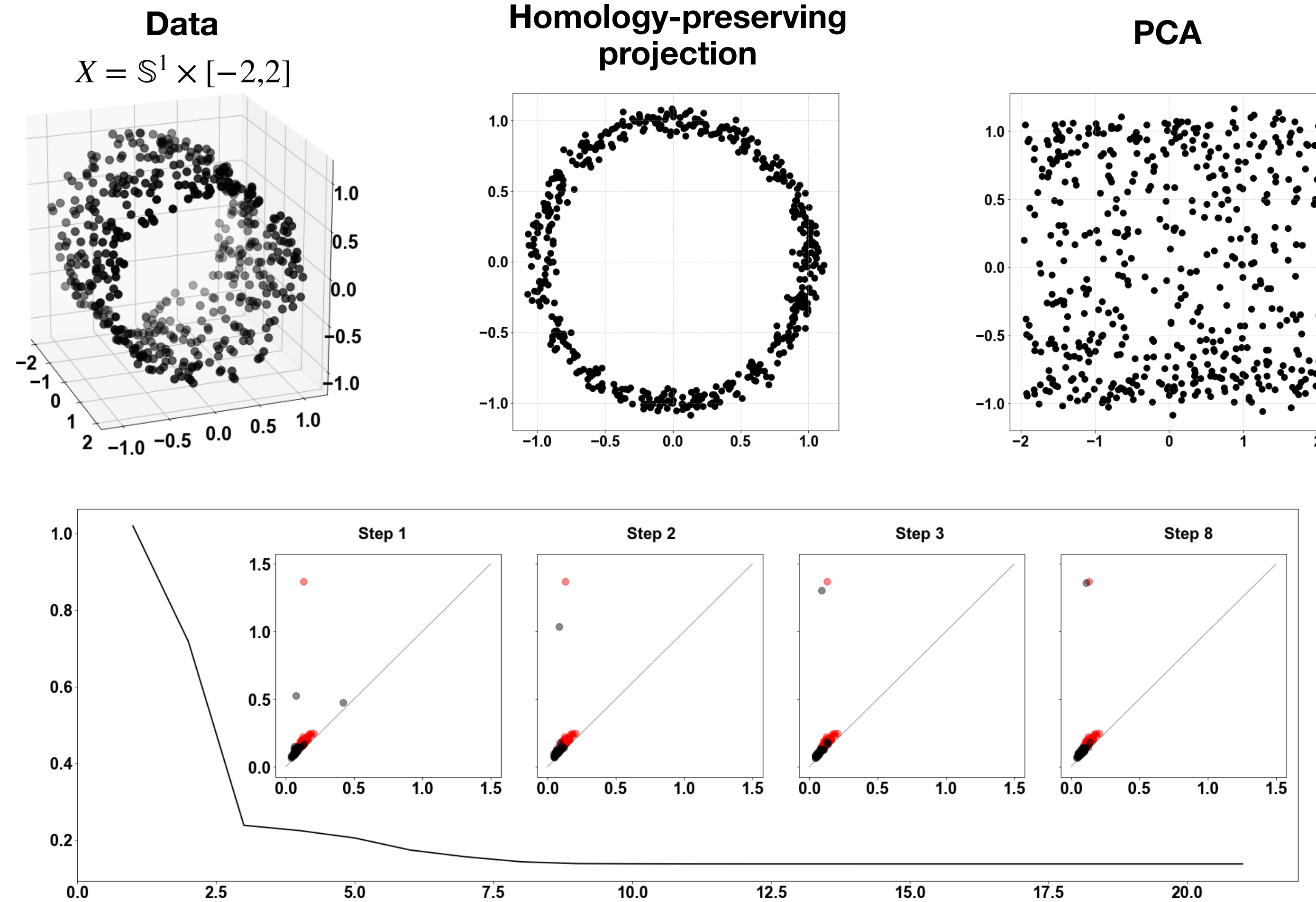
$$(\mathbf{M})_{ij} = d^p(\mathbf{x}_i, \mathbf{y}_j) \quad (\Delta_{D(X)})_i = d^p(\mathbf{x}_i, \partial\Omega) \\ (\Delta_{D(Y)})_j = d^p(\mathbf{y}_j, \partial\Omega)$$



Approximations

- Sinkhorn divergence, $O(n^2)$
- Sliced Wasserstein distance, $O(n \log n)$

Persistent homology-based projection pursuit



Loss function and persistence diagrams of data $D(X)$ (red, fixed) and low-dimensional projection $D(Z)$ (black).

Topological autoencoders

$$L = L_{REC}(X, (h \circ g)(X)) + \lambda L_{TOPO}$$

$$L_{TOPO} = L_{X \rightarrow Z} + L_{Z \rightarrow X}$$

$$L_{X \rightarrow Z} = \|\mathbf{A}_X[\pi_X] - \mathbf{A}_Z[\pi_X]\|^2$$

$$L_{Z \rightarrow X} = \|\mathbf{A}_Z[\pi_Z] - \mathbf{A}_X[\pi_Z]\|^2$$

$$D_X = \{(10,20), (12,30), (15,50)\}$$

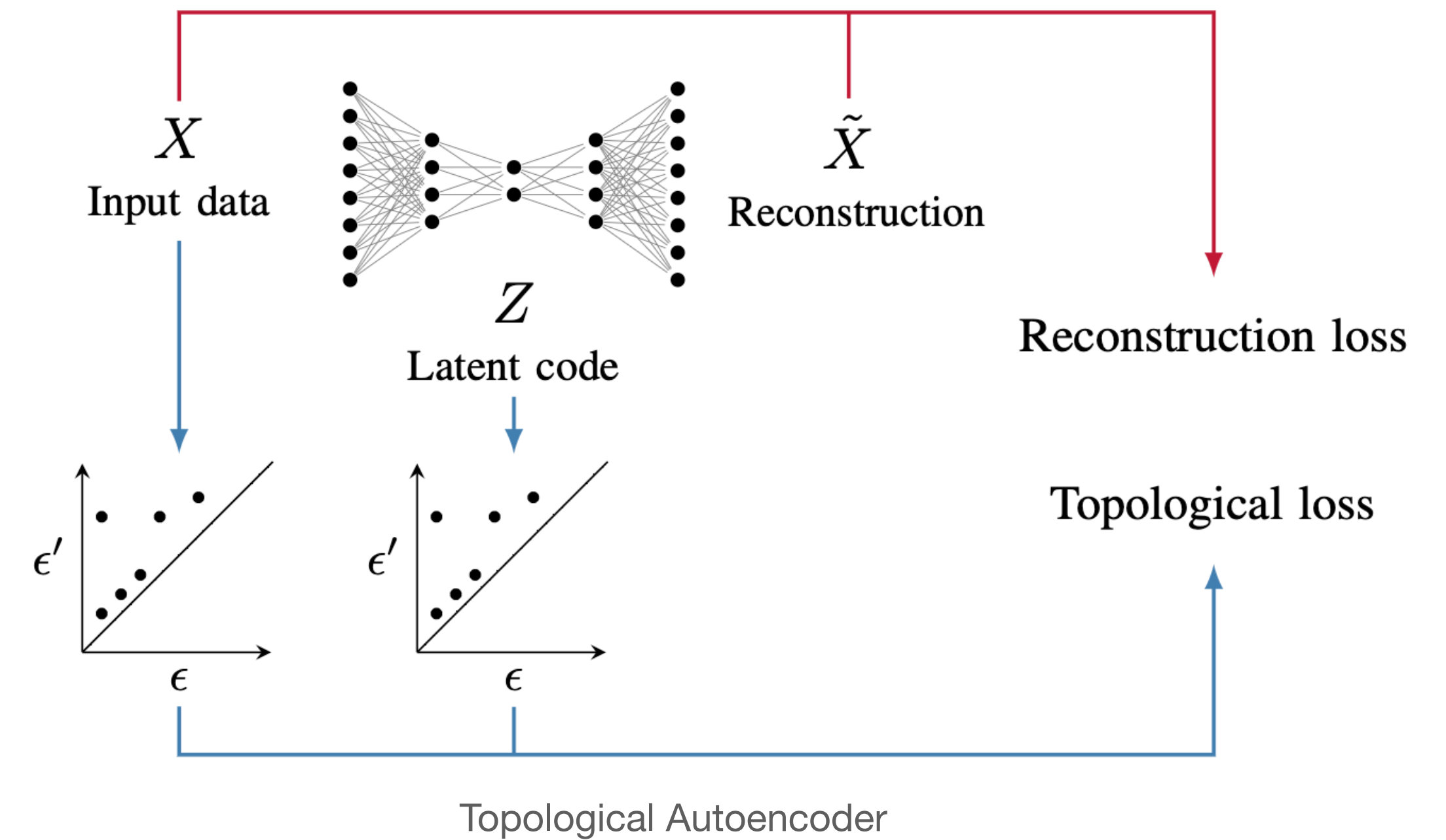
$$\pi_X = \{(1,12), (2,24), (4,34)\}$$

	1	2	3	4
1		20		
2				30
3				50
4				

$\mathbf{A}_X[\pi_X]$

	1	2	3	4
1		2		
2				3
3				5
4				

$\mathbf{A}_Z[\pi_X]$



$$X \rightarrow Z \rightarrow \tilde{X}$$

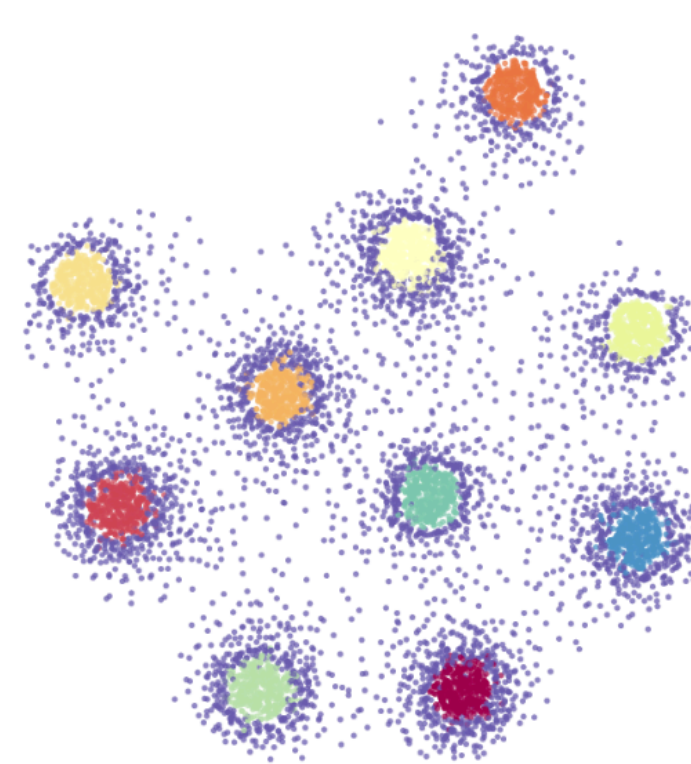
$$Z \cong X$$

Topological autoencoders

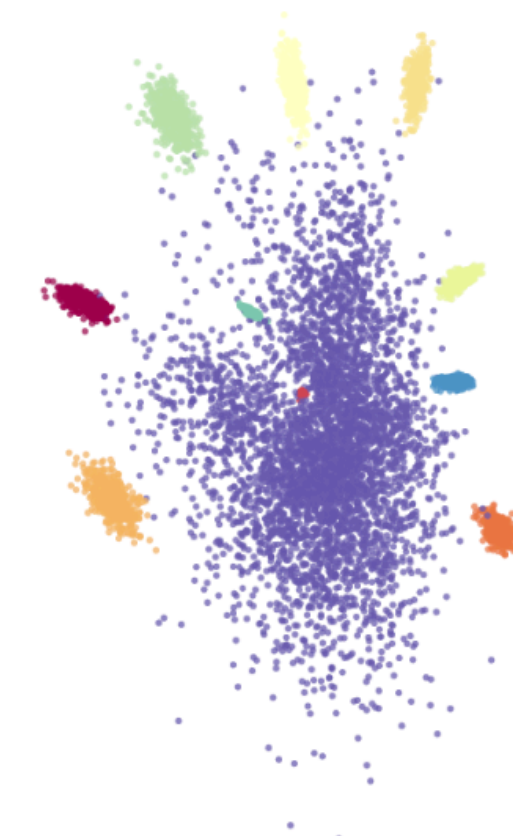
SPHERES data set that consists of ten high-dimensional 100-spheres or radius R living in a 101-dimensional space that are enclosed by one larger sphere of radius $5R$ that consists of the same number of points as the total of inner spheres.



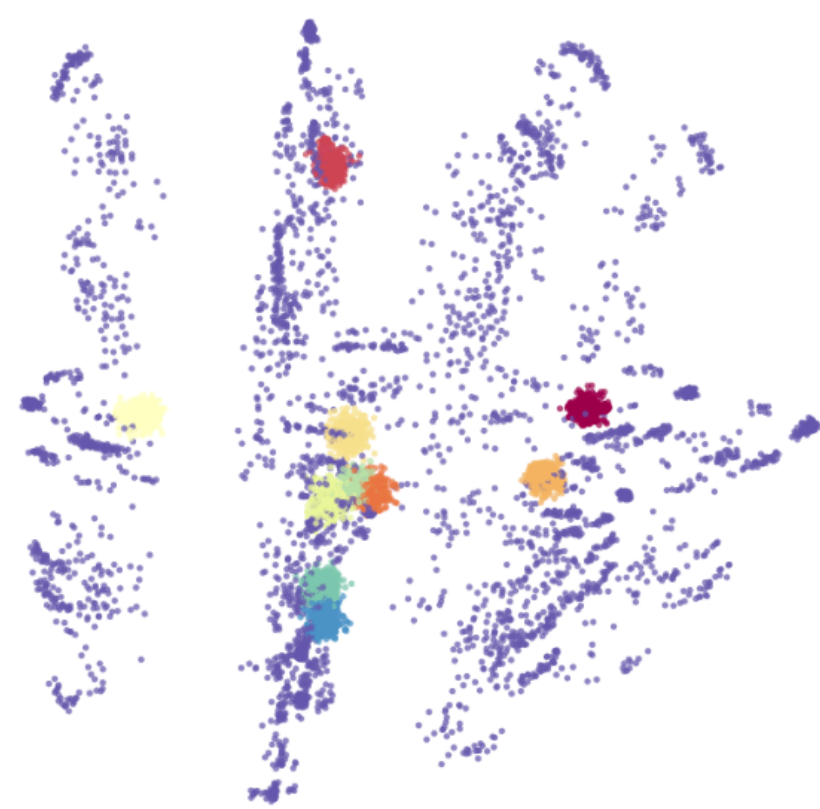
PCA



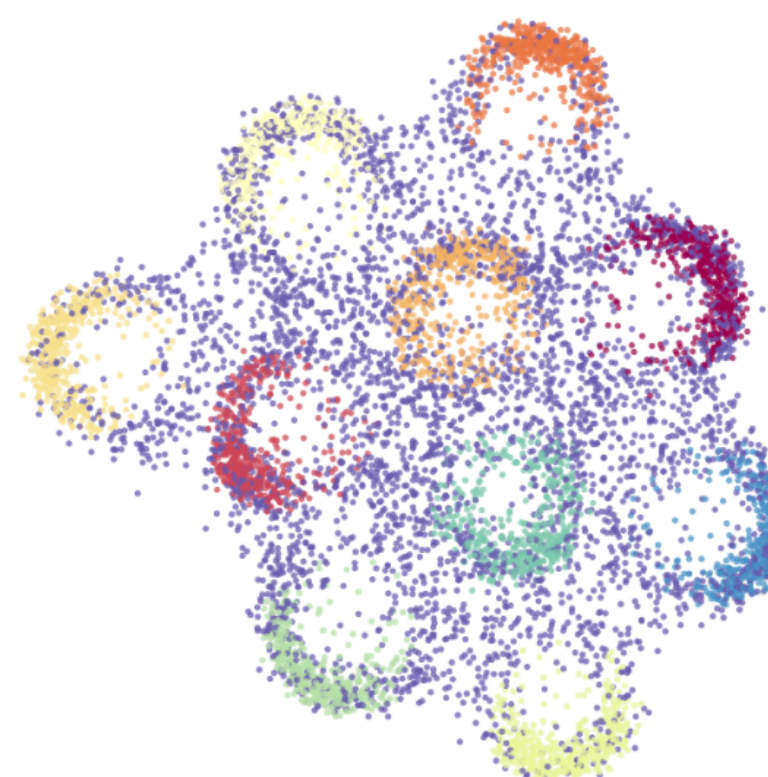
t-SNE



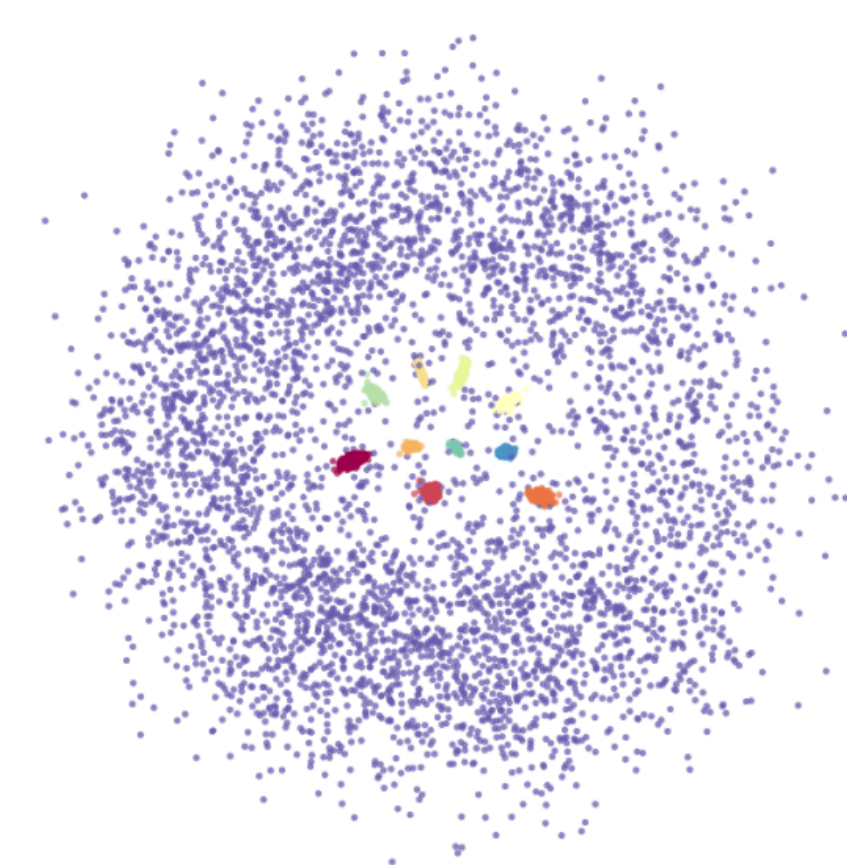
AE



Isomap

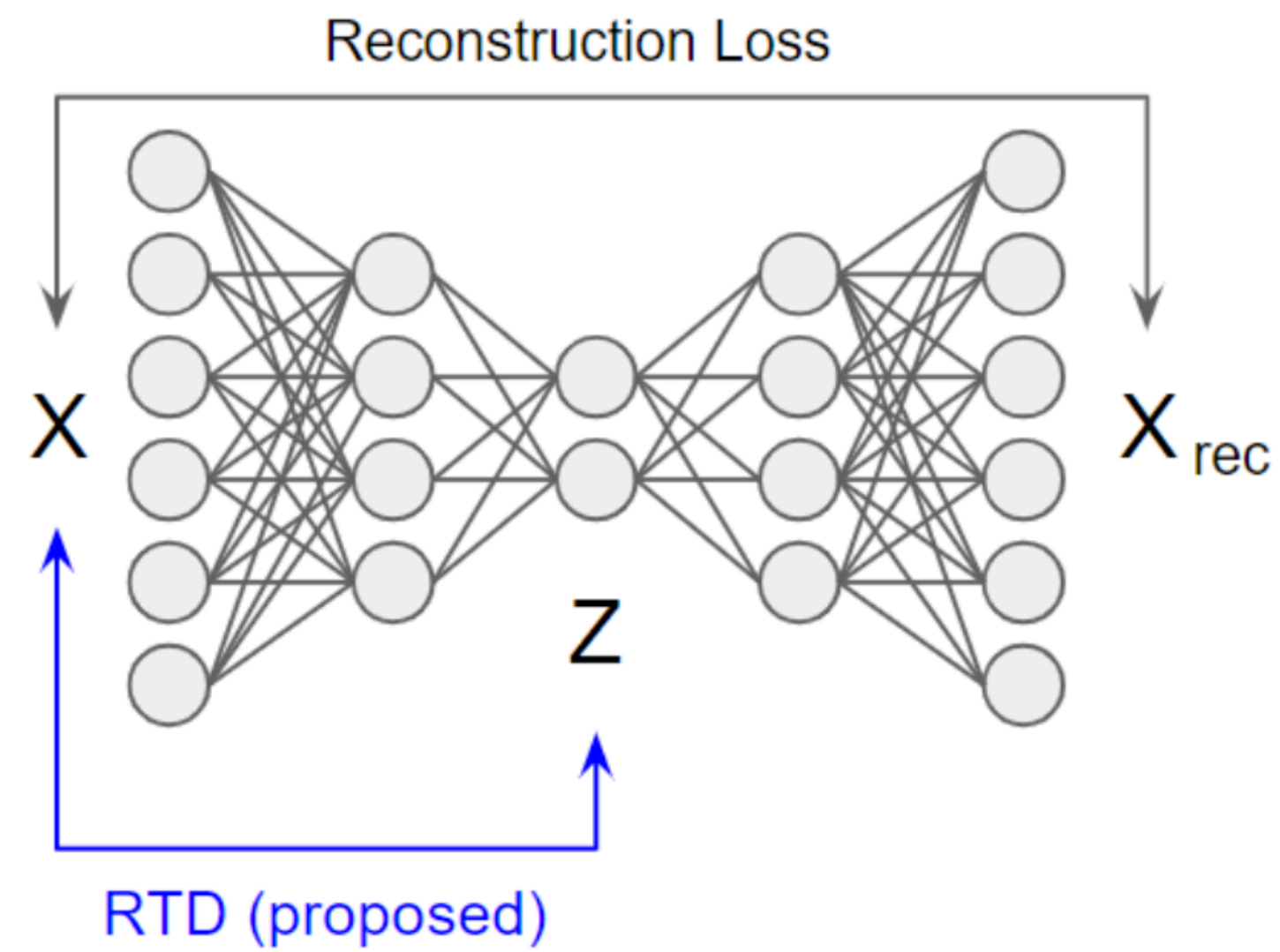


UMAP



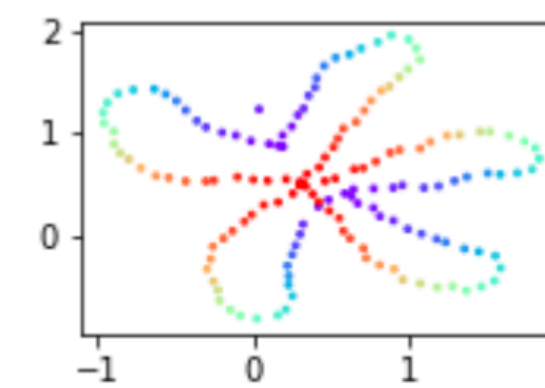
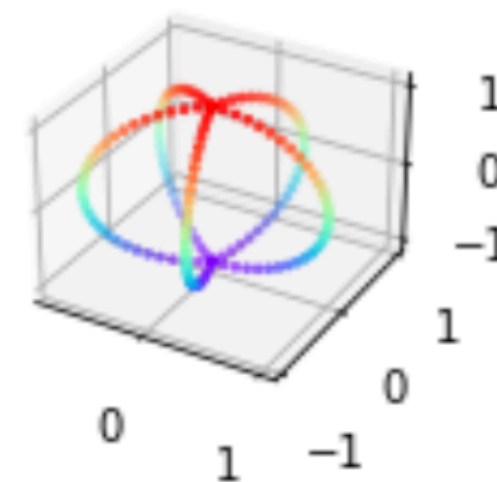
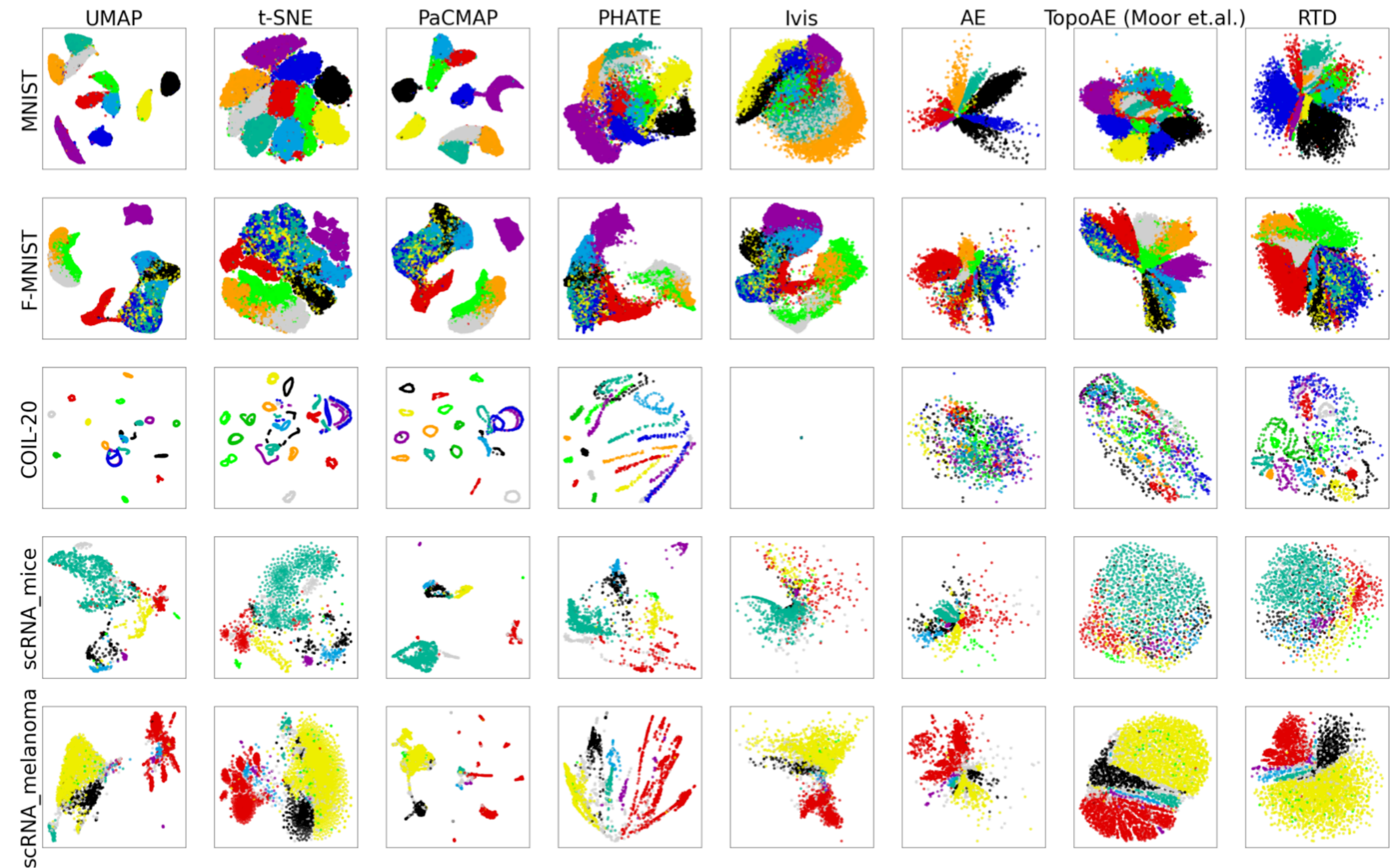
TopoAE

Topology-preserving representation learning

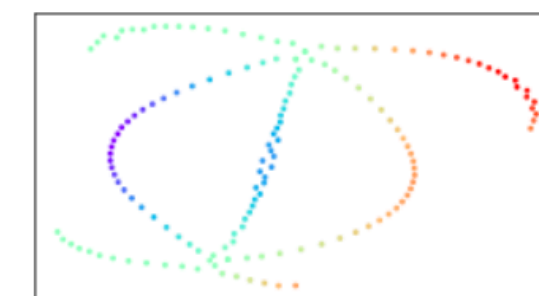


$$X \rightarrow Z \rightarrow \tilde{X}$$

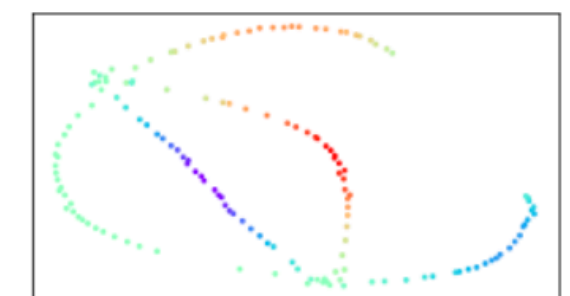
$$Z \cong X$$



(b) RTD

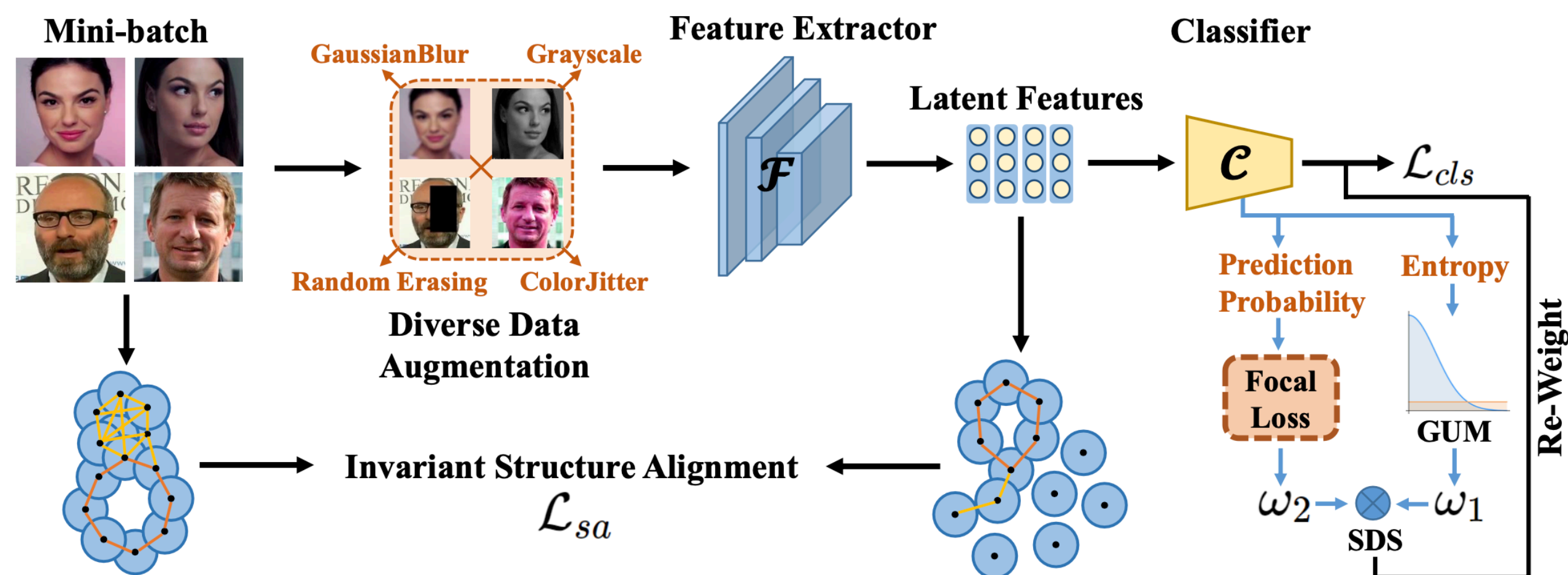


(c) t-SNE



(d) UMAP

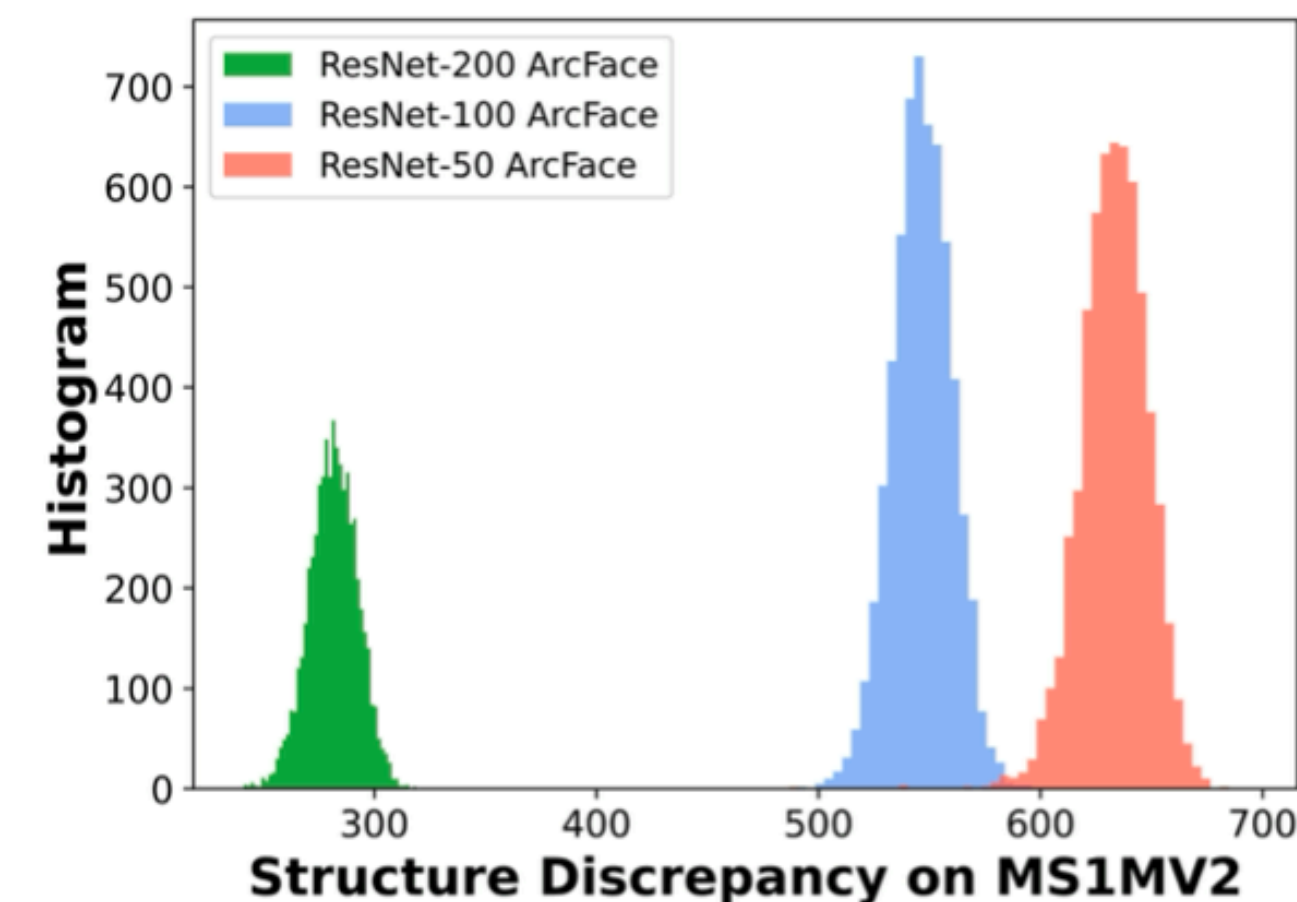
Topology alignment for face recognition



$$L = L_{CLS} + \lambda L_{SA}$$

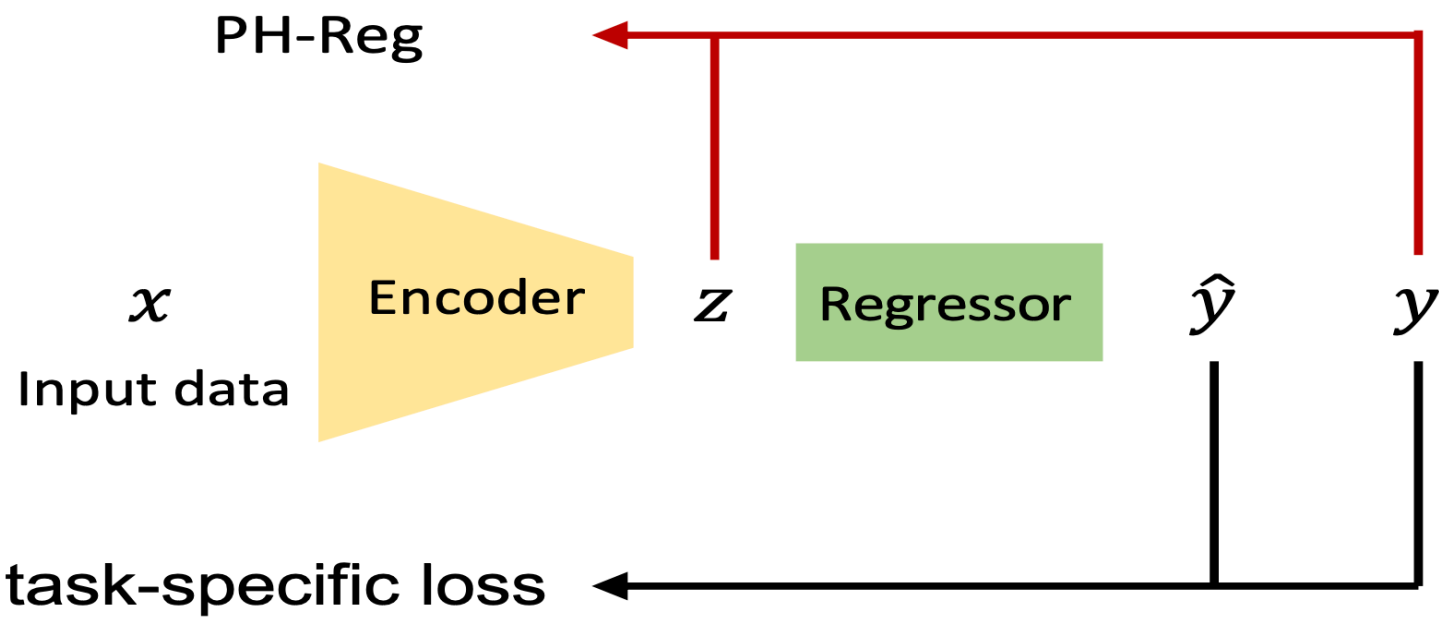
$$X \rightarrow Z \rightarrow Y$$

$$Z \cong X$$



Training Data	Method	IJB-C(1e-6)	IJB-C(1e-5)	IJB-C(1e-4)
MS1MV2	R50, ArcFace Deng et al. (2019)	80.52	88.36	92.52
	R50, MagFace Meng et al. (2021)	81.69	88.95	93.34
	R50, AdaFace Kim et al. (2022)	-	-	96.27
	R50, TopoFR[†]	89.32	94.77	96.40
	R50, TopoFR	90.52	94.71	96.49
	R100, CosFace Wang et al. (2018b)	87.96	92.68	95.56
	R100, ArcFace Deng et al. (2019)	85.65	92.69	95.74
	R100, MV-Softmax Wang et al. (2020)	-	-	95.20
	R100, CircleLoss Sun et al. (2020)	-	89.60	93.95
	R100, URL Shi et al. (2020)	-	95.00	96.60
	R100, BroadFace Kim et al. (2020)	85.96	94.59	96.38
	R100, CurricularFace Huang et al. (2020)	-	-	96.10
	R100, MagFace+ Meng et al. (2021)	90.24	94.08	95.97
	R100, SCF Li et al. (2021b)	-	94.78	96.22
	R100, DAM-CurricularFace Liu et al. (2021)	-	-	96.20
	R100, ElasticFace+ Boutros et al. (2022)	-	-	96.65
	R100, AdaFace Kim et al. (2022)	-	-	96.89
	R100, TopoFR[†]	87.90	95.27	96.90
	R100, TopoFR	90.21	95.23	96.95
	R200, ArcFace Deng et al. (2019)	85.75	94.67	96.53
	R200, TopoFR[†]	87.90	95.19	97.12
	R200, TopoFR	88.39	95.15	97.08
Glint360K	R50, ArcFace Deng et al. (2019)	88.40	95.29	96.81
	R50, TopoFR	91.30	95.99	97.27
	R100, ArcFace Deng et al. (2019)	86.97	96.09	97.04
	R100, TopoFR	91.06	96.57	97.60
	R200, ArcFace Deng et al. (2019)	89.45	95.71	97.20
	R200, TopoFR	86.84	96.54	97.74

Deep regression representation learning with topology



$$X \rightarrow Z \rightarrow Y$$

$$Z \cong Y$$

Table 3: Results on AgeDB

Method	MAE ↓		
	ALL	Many	Med.
Baseline (Yang et al., 2021)	7.77	6.62	9.55
+ $\mathcal{L}'_d$	7.81	6.96	8.88
+ $\mathcal{L}_d$	7.55	6.81	8.43
+ $\mathcal{L}_t$	7.50	6.58	8.79
+ $\mathcal{L}_d + \mathcal{L}_t$	7.48	6.52	8.71

